

EwEMSPPlugin Namespaces

'EwE tools for MSP' is a set of two .NET assemblies that assist Ecopath with Ecosim (EwE) modellers in the task of preparing and testing an Ecospace scenario for game play in the Maritime Strategic Planning (MSP) Challenge 2050 software, and provides all the necessary building blocks to integrate an EwE model into MSP game play.

Namespaces

Namespace	Description
EwEMSPPlugin	Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop
EwEMSPPlugin.Emulator	MEL emulator support for Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop
EwEMSPPlugin.UI	User interface elements for the Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop

EwEMSPPlugin Namespace

Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop

Classes

	Class	Description
	cMSPewEPluginPoint	Plug-in point for the EwE MSP plug-in, integrating the plug-in into • The EwE6 menu • Ecospace initialization events, • Ecospace scenario lifespan events, • Ecospace run preparation events, • The EwE6 help system.

cMSPEwEPluginPoint Class

Plug-in point for the EwE MSP plug-in, integrating the plug-in into

- The EwE6 menu
- Ecospace initialization events,
- Ecospace scenario lifespan events,
- Ecospace run preparation events,
- The EwE6 help system.

Inheritance Hierarchy

[System.Object](#)

EwEMSPPlugin.cMSPEwEPluginPoint

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Class cMSPEwEPluginPoint  
    Implements IMenuItemPlugin, IUIContextPlugin, IEcospacePlugin,  
    IEcospaceInitializedPlugin,  
    IEcospaceInitRunCompletedPlugin, IHelpPlugin
```

The **cMSPEwEPluginPoint** type exposes the following members.

Constructors

	Name	Description
	cMSPEwEPluginPoint	Initializes a new instance of the cMSPEwEPluginPoint class

Properties

	Name	Description
	Author	Plug-in property to report the author of the plug-in.
	Contact	Plug-in property to report contact information about the plug-in.
	ControllImage	Plug-in property to report the control image that the EwE6 UI should show for this plug-in.
	ControlText	Plug-in property to report the control text for this plug-in.
	ControlTooltipText	Plug-in property to report the tool tip text to display this plug-in.
	Description	Plug-in property to describe the plug-in.

 EnabledState	Plug-in property to report the core run state that needs to be met to enable the user interfaces for this plug-in.
 HelpTopic	Plug-in point to report the topic within the help file.
 HelpURL	Plug-in point to report the location of the help file for this plug-in.
 MenuItemLocation	Plug-in property to report the EwE menu item location to place this plug-in.
 Name	Plug-in property to report the unique name of this plug-in. This name is meant to distinguish plug-ins, and will not be visible to users.

Methods

Name	Description
 CloseEcospaceScenario	Plug-in point that is called when an Ecospace scenario has been closed. This plug-in point is used to close the plug-in UI if it exists, and to terminate the EwE shell.
 EcospaceInitialized	Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. The plug-in point is used to store a reference to the Ecospace data structures that will be used while Ecospace is alive.
 EcospaceInitRunCompleted	Plug-in point that is called when Ecospace is about to start running. The plug-in point is needed to set run flags that may have been reset by Ecospace at the end of the last execution.
 Initialize	Plug-in point to initialize to the EwE core. Ignored here in favor of plug-in point UIContext(Object)
 LoadEcospaceScenario	Plug-in point that is called when Ecospace has loaded a scenario, exposing the data source that the scenario was loaded from. This point is needed to initialize an instance of EwE shell against the newly loaded Ecospace data.
 OnControlClick	Plug-in point event handler, that will be called when the control for this plug-in is clicked or activated.
 SaveEcospaceScenario	Plug-in point that is called when Ecospace has saved a scenario, exposing the data source that the scenario was loaded from. The plug-in point is ignored here.
 UIContext	Plug-in point to receive the user interface context to operate onto.

See Also

[EwEMSPPlugin Namespace](#)

IMenuItemPlugin

IUIContextPlugin

IEcospacePlugin

IEcospaceInitializedPlugin

IEcospaceInitRunCompletedPlugin

IHelpPlugin

cMSPEwEPluginPoint Constructor

Initializes a new instance of the [cMSPEwEPluginPoint](#) class

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Sub New

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.cMSPEwEPluginPoint Properties

The [cMSPEwEPluginPoint](#) type exposes the following members.

Properties

	Name	Description
	Author	Plug-in property to report the author of the plug-in.
	Contact	Plug-in property to report contact information about the plug-in.
	ControllImage	Plug-in property to report the control image that the EwE6 UI should show for this plug-in.
	ControlText	Plug-in property to report the control text for this plug-in.
	ControlTooltipText	Plug-in property to report the tool tip text to display this plug-in.
	Description	Plug-in property to describe the plug-in.
	EnabledState	Plug-in property to report the core run state that needs to be met to enable the user interfaces for this plug-in.
	HelpTopic	Plug-in point to report the topic within the help file.
	HelpURL	Plug-in point to report the location of the help file for this plug-in.
	MenuItemLocation	Plug-in property to report the EwE menu item location to place this plug-in.
	Name	Plug-in property to report the unique name of this plug-in. This name is meant to distinguish plug-ins, and will not be visible to users.

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.Author Property

Plug-in property to report the author of the plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Author As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Author

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.Contact Property

Plug-in property to report contact information about the plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Contact As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Contact

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.ControlImage Property

Plug-in property to report the control image that the EwE6 UI should show for this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property ControlImage As Image Get</pre>

Property Value

Type: [Image](#)

Implements

IGUIPlugin.ControlImage

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.ControlText Property

Plug-in property to report the control text for this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property ControlText As String Get</pre>

Property Value

Type: [String](#)

Implements

IGUIPlugin.ControlText

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.ControlTooltipText Property

Plug-in property to report the tool tip text to display this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property ControlTooltipText As String Get</pre>

Property Value

Type: [String](#)

Implements

IGUIPlugin.ControlTooltipText

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.Description Property

Plug-in property to describe the plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Description As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Description

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.EnabledState Property

Plug-in property to report the core run state that needs to be met to enable the user interfaces for this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public ReadOnly Property EnabledState As eCoreExecutionState  
    Get
```

Property Value

Type: **eCoreExecutionState**

Implements

IGUIPlugin.EnabledState

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.HelpTopic Property

Plug-in point to report the topic within the help file.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property HelpTopic As String Get</pre>

Property Value

Type: [String](#)

Implements

IHelpPlugin.HelpTopic

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.HelpURL Property

Plug-in point to report the location of the help file for this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property HelpURL As String Get</pre>

Property Value

Type: [String](#)

Implements

IHelpPlugin.HelpURL

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.MenuItemLocation Property

Plug-in property to report the EwE menu item location to place this plug-in.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property MenuItemLocation As String Get</pre>

Property Value

Type: [String](#)

Implements

IMenuItemPlugin.MenuItemLocation

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.Name Property

Plug-in property to report the unique name of this plug-in. This name is meant to distinguish plug-ins, and will not be visible to users.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public ReadOnly Property Name As String  
    Get
```

Property Value

Type: [String](#)

Implements

IPlugin.Name

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.cMSPEwEPluginPoint Methods

The [cMSPEwEPluginPoint](#) type exposes the following members.

Methods

	Name	Description
	CloseEcospaceScenario	Plug-in point that is called when an Ecospace scenario has been closed. This plug-in point is used to close the plug-in UI if it exists, and to terminate the EwE shell.
	EcospaceInitialized	Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. The plug-in point is used to store a reference to the Ecospace data structures that will be used while Ecospace is alive.
	EcospaceInitRunCompleted	Plug-in point that is called when Ecospace is about to start running. The plug-in point is needed to set run flags that may have been reset by Ecospace at the end of the last execution.
	Initialize	Plug-in point to initialize to the EwE core. Ignored here in favor of plug-in point UIContext(Object)
	LoadEcospaceScenario	Plug-in point that is called when Ecospace has loaded a scenario, exposing the data source that the scenario was loaded from. This point is needed to initialize an instance of EwE shell against the newly loaded Ecospace data.
	OnControlClick	Plug-in point event handler, that will be called when the control for this plug-in is clicked or activated.
	SaveEcospaceScenario	Plug-in point that is called when Ecospace has saved a scenario, exposing the data source that the scenario was loaded from. The plug-in point is ignored here.
	UIContext	Plug-in point to receive the user interface context to operate onto.

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.CloseEcospaceScenario Method

Plug-in point that is called when an Ecospace scenario has been closed. This plug-in point is used to close the plug-in UI if it exists, and to terminate the EwE shell.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub CloseEcospaceScenario
```

Implements

IEcospacePlugin.CloseEcospaceScenario()

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.EcospaceInitialized Method

Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. The plug-in point is used to store a reference to the Ecospace data structures that will be used while Ecospace is alive.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub EcospaceInitialized (  
    EcospaceDatastructures As Object  
)
```

Parameters

EcospaceDatastructures

Type: [System.Object](#)

The ecospace data structures that just received new scenario data.

Implements

IEcospaceInitializedPlugin.EcospaceInitialized(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.EcospaceInitRunCompleted Method

Plug-in point that is called when Ecospace is about to start running. The plug-in point is needed to set run flags that may have been reset by Ecospace at the end of the last execution.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub EcospaceInitRunCompleted (  
    EcospaceDatastructures As Object  
)
```

Parameters

EcospaceDatastructures

Type: [System.Object](#)

The ecospace data structures.

Implements

IEcospaceInitRunCompletedPlugin.EcospaceInitRunCompleted(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.Initialize Method

Plug-in point to initialize to the EwE core. Ignored here in favor of plug-in point [UIContext\(Object\)](#)

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Initialize (  
    core As Object  
)
```

Parameters

core

Type: [System.Object](#)

The core this plug-in is initialized for.

Implements

IPlugin.Initialize(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.LoadEcospaceScenario Method

Plug-in point that is called when Ecospace has loaded a scenario, exposing the data source that the scenario was loaded from. This point is needed to initialize an instance of EwE shell against the newly loaded Ecospace data.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Sub LoadEcospaceScenario (dataSource As Object)</pre>

Parameters

dataSource

Type: [System.Object](#)

A reference to the EwE data source from which data is being loaded.

Implements

IEcospacePlugin.LoadEcospaceScenario(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.OnControlClick Method

Plug-in point event handler, that will be called when the control for this plug-in is clicked or activated.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub OnControlClick (  
    sender As Object,  
    e As EventArgs,  
    ByRef frmPlugin As Form  
)
```

Parameters

sender

Type: [System.Object](#)

The control that was clicked or activated.

e

Type: [System.EventArgs](#)

Event parameters pertaining the control.

frmPlugin

Type: [System.Windows.Forms.Form](#)

A reference to the user interface for this plug-in.

Implements

IGUIPlugin.OnControlClick(Object, EventArgs, Form)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.SaveEcospaceScenario Method

Plug-in point that is called when Ecospace has saved a scenario, exposing the data source that the scenario was loaded from. The plug-in point is ignored here.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub SaveEcospaceScenario (  
    dataSource As Object  
)
```

Parameters

dataSource

Type: [System.Object](#)

A reference to the EwE data source to which data is being saved.

Implements

IEcospacePlugin.SaveEcospaceScenario(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

cMSPEwEPluginPoint.UIContext Method

Plug-in point to receive the **user interface context** to operate onto.

Namespace: [EwEMSPPlugin](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub UIContext (  
    uic As Object  
)
```

Parameters

uic

Type: [System.Object](#)

The **user interface context** to operate onto

Implements

IUIContextPlugin.UIContext(Object)

See Also

[cMSPEwEPluginPoint Class](#)

[EwEMSPPlugin Namespace](#)

EwEMSPPlugin.Emulator Namespace

MEL emulator support for Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop

Classes

	Class	Description
	cTestset	Set of test input for configured pressures >
	cTestSetData	Container for cTestset instances.

cTestset Class

Set of test input for configured [pressures](#)>

Inheritance Hierarchy

[System.Object](#)

EwEMSPPlugin.Emulator.cTestset

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cTestset Implements IMELItem</pre>

The **cTestset** type exposes the following members.

Constructors

	Name	Description
	cTestset	Create a new cTestset .

Properties

	Name	Description
	Name	Get/set the name of the testset.
	Pressures	Obtain the pressures for which testdata is defined.
	Testdata	Get/set the testdata for a given pressure

Methods

	Name	Description
	ToString	Formatter, displays the name of the testset. (Overrides Object.ToString() .)

See Also

[EwEMSPPlugin.Emulator Namespace](#)

[EwEShell.IMELItem](#)

cTestset Constructor

Create a new [cTestset](#).

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    strName As String,  
    g As cGame  
)
```

Parameters

strName

Type: [System.String](#)

Name of the test set.

g

Type: [EwEShell.cGame](#)

The [game](#) to define the testset for.

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.cTestset Properties

The [cTestset](#) type exposes the following members.

Properties

	Name	Description
	Name	Get/set the name of the testset.
	Pressures	Obtain the pressures for which testdata is defined.
	Testdata	Get/set the testdata for a given pressure

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.Name Property

Get/set the name of the testset.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.Pressures Property

Obtain the pressures for which testdata is defined.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Pressures As ICollection(Of cPressure) Get</pre>

Property Value

Type: [ICollection\(cPressure\)](#)

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.Testdata Property

Get/set the testdata for a given [pressure](#)

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
  
Public Property Testdata (  
    pressure As cPressure  
) As String  
    Get  
    Set
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

Property Value

Type: [String](#)

The testdata.

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.cTestset Methods

The [cTestset](#) type exposes the following members.

Methods

	Name	Description
	ToString	Formatter, displays the name of the testset. (Overrides Object.ToString() .)

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestset.ToString Method

Formatter, displays the name of the testset.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ToString As String</code>

Return Value

Type: [String](#)

A [String](#) that represents the testset

See Also

[cTestset Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData Class

Container for [cTestset](#) instances.

Inheritance Hierarchy

[System.Object](#)

EwEMSPPlugin.Emulator.cTestSetData

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cTestSetData</pre>

The **cTestSetData** type exposes the following members.

Constructors

	Name	Description
	cTestSetData	Create new cTestSetData .

Properties

	Name	Description
	Testsets	Returns the list of testsets that are currently available.

Methods

	Name	Description
	Close	Closes this instance and discard all testset information.
	Load	Loads all available testsets from persistent storage compatible with a given EwE model, ecospace scenario and MSP game.
	Save	Saves testsets back to persistent storage.

See Also

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData Constructor

Create new [cTestSetData](#).

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Sub New

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.cTestSetData Properties

The [cTestSetData](#) type exposes the following members.

Properties

	Name	Description
	Testsets	Returns the list of testsets that are currently available.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.Testsets Property

Returns the list of testsets that are currently available.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Testsets As List(Of cTestset) Get</pre>

Return Value

Type: [List\(cTestset\)](#)

The list of testsets that are currently available.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.cTestSetData Methods

The [cTestSetData](#) type exposes the following members.

Methods

	Name	Description
	Close	Closes this instance and discard all testset information.
	Load	Loads all available testsets from persistent storage compatible with a given EwE model, ecospace scenario and MSP game.
	Save	Saves testsets back to persistent storage.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.Close Method

Closes this instance and discard all testset information.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function Close As Boolean

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.Load Method

Loads all available testsets from persistent storage compatible with a given EwE model, ecospace scenario and MSP game.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function Load (  
    strModelScenario As String,  
    g As cGame  
) As Boolean
```

Parameters

strModelScenario

Type: [System.String](#)

The string model scenario.

g

Type: [EwEShell.cGame](#)

The g.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

cTestSetData.Save Method

Saves testsets back to persistent storage.

Namespace: [EwEMSPPlugin.Emulator](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function Save As Boolean

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cTestSetData Class](#)

[EwEMSPPlugin.Emulator Namespace](#)

EwEMSPPlugin.UI Namespace

User interface elements for the Maritime Strategic Planning Challenge 2050 plug-in for EwE desktop

Classes

	Class	Description
	cEcosimFishingChecker	Helper class to validate if Ecosim contains conflicting fishing patterns. These patterns will need removing from Ecosim, as fisheries dynamics will be replaced by MSP player actions.
	cEcosimForcingChecker	Helper class to validate if Ecosim contains conflicting forcing patterns among applied forcing functions. These patterns will need removing from Ecosim, as temporal forcing dynamics will be replaced by MSP player actions.
	cEcosimTimeSeriesChecker	Helper class to validate if Ecosim contains loaded time series. These patterns will need removing from Ecosim, as MSP will not run in an absolute time.
	cEcospaceTimeSeriesChecker	Helper class to validate if Ecospace is forced with spatial-temporal data. Any forcing patterns will need removing from Ecospace as they will be replaced with MSP player interactions.
	cRequirementChecker	Base class for validating aspects of the EwE model for suitability for MSP gameplay.
	frmGameDesigner	Main user interface for the MSP tools for EwE desktop plug-in.
	gridDrivers	Driver configuration grid; enables configuring how MEL pressures map to available Ecospace driver variables .
	gridEmulator	Test data configuration grid; enables managing simulated MEL test data .
	gridOutcomes	Outcome configuration grid; enables configuring how Ecospace predictions are fed back to MEL.

Delegates

	Delegate	Description
	gridDrivers.OnMappingsChangedEventHandler	
	gridEmulator.OnTestsetChangedEventHandler	
	gridOutcomes.OnMappingsChangedEventHandler	

cEcosimFishingChecker Class

Helper class to validate if Ecosim contains conflicting fishing patterns. These patterns will need removing from Ecosim, as fisheries dynamics will be replaced by MSP player actions.

Inheritance Hierarchy

[System.Object](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

EwEMSPPlugin.UI.cEcosimFishingChecker

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEcosimFishingChecker Inherits cRequirementChecker</pre>

The **cEcosimFishingChecker** type exposes the following members.

Constructors

	Name	Description
	cEcosimFishingChecker	Initializes a new instance of the cEcosimFishingChecker class.

Methods

	Name	Description
	CheckRequirements	The requirement check for conflicting Ecosim fishing patterns, both for fishing effort and fishing mortality. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim fishing dynamics. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[EwEMSPPlugin.UI Namespace](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

cEcosimFishingChecker Constructor

Initializes a new instance of the [cEcosimFishingChecker](#) class.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core containing Ecosim to validate.

See Also

[cEcosimFishingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimFishingChecker.cEcosimFishingChecker Methods

The [cEcosimFishingChecker](#) type exposes the following members.

Methods

	Name	Description
	CheckRequirements	The requirement check for conflicting Ecosim fishing patterns, both for fishing effort and fishing mortality. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim fishing dynamics. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[cEcosimFishingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimFishingChecker.CheckRequirements Method

The requirement check for conflicting Ecosim fishing patterns, both for fishing effort and fishing mortality.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected Overrides Sub CheckRequirements

See Also

[cEcosimFishingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimFishingChecker.OnCoreMessage Method

Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim fishing dynamics.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Sub OnCoreMessage (  
    msg As cMessage  
)
```

Parameters

msg

Type: **cMessage**

The message to respond to.

See Also

[cEcosimFishingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimForcingChecker Class

Helper class to validate if Ecosim contains conflicting forcing patterns among applied forcing functions. These patterns will need removing from Ecosim, as temporal forcing dynamics will be replaced by MSP player actions.

Inheritance Hierarchy

[System.Object](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

EwEMSPPlugin.UI.cEcosimForcingChecker

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEcosimForcingChecker Inherits cRequirementChecker</pre>

The **cEcosimForcingChecker** type exposes the following members.

Constructors

	Name	Description
	cEcosimForcingChecker	Initializes a new instance of the cEcosimForcingChecker class.

Methods

	Name	Description
	CheckRequirements	The requirement check for existing forcing patterns among applied forcing functions. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim forcing functions. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

Remarks

This class still needs to consider applied environmental forcing functions.

See Also

[EwEMSPPlugin.UI Namespace](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

cEcosimForcingChecker Constructor

Initializes a new instance of the [cEcosimForcingChecker](#) class.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core containing Ecosim to validate.

See Also

[cEcosimForcingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimForcingChecker.cEcosimForcingChecker Methods

The [cEcosimForcingChecker](#) type exposes the following members.

Methods

	Name	Description
	CheckRequirements	The requirement check for existing forcing patterns among applied forcing functions. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim forcing functions. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[cEcosimForcingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimForcingChecker.CheckRequirements Method

The requirement check for existing forcing patterns among applied forcing functions.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected Overrides Sub CheckRequirements

See Also

[cEcosimForcingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimForcingChecker.OnCoreMessage Method

Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosim forcing functions.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Sub OnCoreMessage (  
    msg As cMessage  
)
```

Parameters

msg

Type: **cMessage**

The message to respond to.

See Also

[cEcosimForcingChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimTimeSeriesChecker Class

Helper class to validate if Ecosim contains loaded time series. These patterns will need removing from Ecosim, as MSP will not run in an absolute time.

Inheritance Hierarchy

[System.Object](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

EwEMSPPlugin.UI.cEcosimTimeSeriesChecker

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEcosimTimeSeriesChecker Inherits cRequirementChecker</pre>

The **cEcosimTimeSeriesChecker** type exposes the following members.

Constructors

	Name	Description
	cEcosimTimeSeriesChecker	Initializes a new instance of the cEcosimTimeSeriesChecker class.

Methods

	Name	Description
	CheckRequirements	The requirement check for loaded Ecosim time series. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosimn time series. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[EwEMSPPlugin.UI Namespace](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

cEcosimTimeSeriesChecker Constructor

Initializes a new instance of the [cEcosimTimeSeriesChecker](#) class.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core containing Ecosim to validate.

See Also

[cEcosimTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimTimeSeriesChecker.cEcosimTimeSeriesChecker Methods

The [cEcosimTimeSeriesChecker](#) type exposes the following members.

Methods

	Name	Description
	CheckRequirements	The requirement check for loaded Ecosim time series. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosimn time series. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[cEcosimTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimTimeSeriesChecker.CheckRequirements Method

The requirement check for loaded Ecosim time series.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected Overrides Sub CheckRequirements

See Also

[cEcosimTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcosimTimeSeriesChecker.OnCoreMessage Method

Core message handler, implemented to automatically trigger a requirement check when the user changes Ecosimn time series.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Sub OnCoreMessage (  
    msg As cMessage  
)
```

Parameters

msg

Type: **cMessage**

The message to respond to.

See Also

[cEcosimTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcospaceTimeSeriesChecker Class

Helper class to validate if Ecospace is forced with spatial-temporal data. Any forcing patterns will need removing from Ecospace as they will be replaced with MSP player interactions.

Inheritance Hierarchy

[System.Object](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

EwEMSPPlugin.UI.cEcospaceTimeSeriesChecker

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Class cEcospaceTimeSeriesChecker  
    Inherits cRequirementChecker
```

The **cEcospaceTimeSeriesChecker** type exposes the following members.

Constructors

	Name	Description
	cEcospaceTimeSeriesChecker	Initializes a new instance of the cEcospaceTimeSeriesChecker class.

Methods

	Name	Description
	CheckRequirements	The requirement check for applied Ecospace external data connections. There should be none. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecospace time series. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[EwEMSPPlugin.UI Namespace](#)

[EwEMSPPlugin.UI.cRequirementChecker](#)

cEcospaceTimeSeriesChecker Constructor

Initializes a new instance of the [cEcospaceTimeSeriesChecker](#) class.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core containing Ecosim to validate.

See Also

[cEcospaceTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcospaceTimeSeriesChecker.cEcospaceTimeSeriesChecker Methods

The [cEcospaceTimeSeriesChecker](#) type exposes the following members.

Methods

	Name	Description
	CheckRequirements	The requirement check for applied Ecospace external data connections. There should be none. (Overrides cRequirementChecker.CheckRequirements() .)
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes Ecospace time series. (Overrides cRequirementChecker.OnCoreMessage(cMessage) .)

See Also

[cEcospaceTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcospaceTimeSeriesChecker.CheckRequirements Method

The requirement check for applied Ecospace external data connections. There should be none.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Protected Overrides Sub CheckRequirements</code>

See Also

[cEcospaceTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cEcospaceTimeSeriesChecker.OnCoreMessage Method

Core message handler, implemented to automatically trigger a requirement check when the user changes Ecospace time series.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides Sub OnCoreMessage (  
    msg As cMessage  
)
```

Parameters

msg

Type: **cMessage**

The message to respond to.

See Also

[cEcospaceTimeSeriesChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker Class

Base class for validating aspects of the EwE model for suitability for MSP gameplay.

Inheritance Hierarchy

[System.Object](#)

EwEMSPPlugin.UI.cRequirementChecker

[EwEMSPPlugin.UI.cEcosimFishingChecker](#)

[EwEMSPPlugin.UI.cEcosimForcingChecker](#)

[EwEMSPPlugin.UI.cEcosimTimeSeriesChecker](#)

[EwEMSPPlugin.UI.cEcospaceTimeSeriesChecker](#)

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public MustInherit Class cRequirementChecker
```

The **cRequirementChecker** type exposes the following members.

Constructors

	Name	Description
	cRequirementChecker	Initializes a new instance of the cRequirementChecker class.

Properties

	Name	Description
	RequirementsMet	Get/set whether the requirements are met.

Methods

	Name	Description
	CheckRequirements	The validation logic, which should set the value of RequirementsMet .
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes the relevant configuration of EwE.

Fields

	Name	Description
	m_core	The cCore instance to check.

See Also

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker Constructor

Initializes a new instance of the [cRequirementChecker](#) class.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core to check.

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.cRequirementChecker Properties

The [cRequirementChecker](#) type exposes the following members.

Properties

	Name	Description
	RequirementsMet	Get/set whether the requirements are met.

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.RequirementsMet Property

Get/set whether the requirements are met.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property RequirementsMet As Boolean Get Protected Set</pre>

Property Value

Type: [Boolean](#)

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.cRequirementChecker Methods

The [cRequirementChecker](#) type exposes the following members.

Methods

	Name	Description
	CheckRequirements	The validation logic, which should set the value of RequirementsMet .
	OnCoreMessage	Core message handler, implemented to automatically trigger a requirement check when the user changes the relevant configuration of EwE.

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.CheckRequirements Method

The validation logic, which should set the value of [RequirementsMet](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Protected MustOverride Sub CheckRequirements</code>

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.OnCoreMessage Method

Core message handler, implemented to automatically trigger a requirement check when the user changes the relevant configuration of EwE.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public MustOverride Sub OnCoreMessage (  
    msg As cMessage  
)
```

Parameters

msg

Type: **cMessage**

The message to respond to.

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.cRequirementChecker Fields

The [cRequirementChecker](#) type exposes the following members.

Fields

	Name	Description
	m_core	The cCore instance to check.

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

cRequirementChecker.m_core Field

The **cCore** instance to check.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected m_core **As** cCore

Field Value

Type: **cCore**

See Also

[cRequirementChecker Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner Class

Main user interface for the MSP tools for EwE desktop plug-in.

Inheritance Hierarchy

[System.Object](#)

[System.MarshalByRefObject](#)

[System.ComponentModel.Component](#)

[System.Windows.Forms.Control](#)

[System.Windows.Forms.ScrollableControl](#)

[System.Windows.Forms.ContainerControl](#)

[System.Windows.Forms.Form](#)

DockContent

frmEwEDockContent

frmEwE

EwEMSPPlugin.UI.frmGameDesigner

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Class frmGameDesigner  
    Inherits frmEwE
```

The **frmGameDesigner** type exposes the following members.

Constructors

	Name	Description
	frmGameDesigner	Creates a new frmGameDesigner .

Properties

	Name	Description
	IsRunForm	overridden to ensure that this form is treated as a run form. Run forms do not close when input data has changed. (Overrides frmEwE.IsRunForm .)

Methods

	Name	Description
	Dispose	Releases the unmanaged resources used by the frmGameDesigner and optionally releases the managed resources (Overrides Form.Dispose(Boolean) .)
	OnCoreMessage	Core message handler to make the UI respond to important EwE events. (Overrides frmEwE.OnCoreMessage(cMessage) .)

 OnFormClosed	Form close event override, cleans up the UI. (Overrides frmEwE.OnFormClosed(FormClosedEventArgs).)
 OnLoad	Form load event override to initialize content. (Overrides frmEwE.OnLoad(EventArgs).)
 SelectedGame	Returns the selected game .
 UpdateControls	Overridden to update the state of controls in this form. (Overrides frmEwE.UpdateControls().)

See Also

[EwEMSPPlugin.UI Namespace](#)

frmEwE

frmGameDesigner Constructor

Creates a new [frmGameDesigner](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    uic As cUIContext,  
    shell As cEwEShell,  
    data As cEcospaceDataStructures  
)
```

Parameters

uic

Type: **cUIContext**

The **cUIContext** to work against.

shell

Type: [EwEShell.cEwEShell](#)

The [cEwEShell](#) that contains MSP game info.

data

Type: **cEcospaceDataStructures**

The **cEcospaceDataStructures** to work against.

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.frmGameDesigner Properties

The [frmGameDesigner](#) type exposes the following members.

Properties

	Name	Description
	IsRunForm	overridden to ensure that this form is treated as a run form. Run forms do not close when input data has changed. (Overrides frmEwE.IsRunForm .)

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.IsRunForm Property

overridden to ensure that this form is treated as a run form. Run forms do not close when input data has changed.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides ReadOnly Property IsRunForm As Boolean  
    Get
```

Property Value

Type: [Boolean](#)

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.frmGameDesigner Methods

The [frmGameDesigner](#) type exposes the following members.

Methods

	Name	Description
	Dispose	(Overrides Form.Dispose(Boolean) .)
	OnCoreMessage	Core message handler to make the UI respond to important EwE events. (Overrides frmEwE.OnCoreMessage(cMessage) .)
	OnFormClosed	Form close event override, cleans up the UI. (Overrides frmEwE.OnFormClosed(FormClosedEventArgs) .)
	OnLoad	Form load event override to initialize content. (Overrides frmEwE.OnLoad(EventArgs) .)
	SelectedGame	Returns the selected game .
	UpdateControls	Overridden to update the state of controls in this form. (Overrides frmEwE.UpdateControls() .)

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.Dispose Method

Releases the unmanaged resources used by the [frmGameDesigner](#) and optionally releases the managed resources

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Overrides Sub Dispose (  
    disposing As Boolean  
)
```

Parameters

disposing

Type: [System.Boolean](#)

True to release both managed and unmanaged resources; false to release only unmanaged resources

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.OnCoreMessage Method

Core message handler to make the UI respond to important EwE events.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Overrides Sub OnCoreMessage (msg As cMessage)</pre>

Parameters

msg

Type: **cMessage**

The subscribed message to respond to.

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.OnFormClosed Method

Form close event override, cleans up the UI.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Overrides Sub OnFormClosed (  
    e As FormClosedEventArgs  
)
```

Parameters

e

Type: [System.Windows.Forms.FormClosedEventArgs](#)

Ignored

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.OnLoad Method

Form load event override to initialize content.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Overrides Sub OnLoad (  
    e As EventArgs  
)
```

Parameters

e

Type: [System.EventArgs](#)

Ignored

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.SelectedGame Method

Returns the selected [game](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function SelectedGame As cGame
```

Return Value

Type: [cGame](#)

The selected [game](#), or Nothing if no game is selected.

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

frmGameDesigner.UpdateControls Method

Overridden to update the state of controls in this form.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub UpdateControls

See Also

[frmGameDesigner Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers Class

Driver configuration grid; enables configuring how MEL [pressures](#) map to available [Ecospace driver variables](#).

Inheritance Hierarchy

[System.Object](#)

[System.MarshalByRefObject](#)

[System.ComponentModel.Component](#)

[System.Windows.Forms.Control](#)

[System.Windows.Forms.ScrollableControl](#)

[System.Windows.Forms.Panel](#)

CustomScrollControl

GridVirtual

Grid

EwEGrid

EwEMSPPlugin.UI.gridDrivers

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Class gridDrivers
    Inherits EwEGrid
```

The **gridDrivers** type exposes the following members.

Constructors

	Name	Description
	gridDrivers	Creates a new driver configuration grid .

Properties

	Name	Description
	Game	Get/set the active game to operate onto.
	SelectedPressure	Get the pressure currently selected in the grid.
	Shell	Get/set the MSP shell to operate onto.

Methods

	Name	Description
	FillData	Fill the grid with pressure - driver mappings. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)

	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update driver mappings. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual) .)

Events

	Name	Description
	OnMappingsChanged	Event; fired when the user has changed a cPressure to cDriver mapping displayed in the grid.

See Also

[EwEMSPPlugin.UI Namespace](#)

EwEGrid

gridDrivers Constructor

Creates a new [driver configuration grid](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB		
Public	Sub	New

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.gridDrivers Properties

The [gridDrivers](#) type exposes the following members.

Properties

	Name	Description
	Game	Get/set the active game to operate onto.
	SelectedPressure	Get the pressure currently selected in the grid.
	Shell	Get/set the MSP shell to operate onto.

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.Game Property

Get/set the active [game](#) to operate onto.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Game As cGame Get Set</pre>

Property Value

Type: [cGame](#)

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.SelectedPressure Property

Get the [pressure](#) currently selected in the grid.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property SelectedPressure As cPressure Get</pre>

Property Value

Type: [cPressure](#)

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.Shell Property

Get/set the [MSP shell](#) to operate onto.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Shell As cEwEShell Get Set</pre>

Property Value

Type: [cEwEShell](#)

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.gridDrivers Methods

The [gridDrivers](#) type exposes the following members.

Methods

	Name	Description
	FillData	Fill the grid with pressure - driver mappings. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)
	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update driver mappings. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual) .)

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.FillData Method

Fill the grid with pressure - driver mappings.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub FillData

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.FinishStyle Method

Finalizes grid formatting.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Protected Overrides Sub FinishStyle</code>

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.InitStyle Method

Initialize the grid columns and layout.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub InitStyle

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.OnCellValueChanged Method

EwEEditHandler

callback for responding to user cell value changes. Overridden to update driver mappings.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Protected Overrides Function OnCellValueChanged (  
    p As Position,  
    cell As ICellVirtual  
) As Boolean
```

Parameters

p

Type: **Position**

Position that was affected.

cell

Type: **ICellVirtual**

Cell that has received a new value.

Return Value

Type: [Boolean](#)

The return value is ignored by the EwEGrid framework.

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.gridDrivers Events

The [gridDrivers](#) type exposes the following members.

Events

	Name	Description
	OnMappingsChanged	Event; fired when the user has changed a cPressure to cDriver mapping displayed in the grid.

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.OnMappingsChanged Event

Event; fired when the user has changed a [cPressure](#) to [cDriver](#) mapping displayed in the grid.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Event OnMappingsChanged As gridDrivers.OnMappingsChangedEventHandler

Value

Type: [EwEMSPPlugin.UI.gridDrivers.OnMappingsChangedEventHandler](#)

See Also

[gridDrivers Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridDrivers.OnMappingsChangedEventHandler Delegate

[Missing <summary> documentation for

"T:EwEMSPPlugin.UI.gridDrivers.OnMappingsChangedEventHandler"]

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Delegate Sub OnMappingsChangedEventHandler (  
    sender As gridDrivers  
)
```

Parameters

sender

Type: [EwEMSPPlugin.UI.gridDrivers](#)

See Also

[EwEMSPPlugin.UI Namespace](#)

gridEmulator Class

Test data configuration grid; enables managing [simulated MEL test data](#).

Inheritance Hierarchy

[System.Object](#)

[System.MarshalByRefObject](#)

[System.ComponentModel.Component](#)

[System.Windows.Forms.Control](#)

[System.Windows.Forms.ScrollableControl](#)

[System.Windows.Forms.Panel](#)

CustomScrollControl

GridVirtual

Grid

EwEGrid

EwEMSPPlugin.UI.gridEmulator

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class gridEmulator Inherits EwEGrid</pre>

The **gridEmulator** type exposes the following members.

Constructors

	Name	Description
	gridEmulator	Creates a new test set configuration .

Properties

	Name	Description
	Game	Get/set the active game to operate onto.
	Pressure	Get/set the pressure associated with a given grid row.
	Testset	Get/set the active test set to operate onto.

Methods

	Name	Description
	FillData	Fill the grid with testset data. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)

	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellClicked	Edit handler to catch driver cell clicks. Overridden to invoke a folder browse dialog when the user clicks a Grid pressure cell. (Overrides EwEGrid.OnCellClicked(Position, ICellVirtual).)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update testset content. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual).)

Events

	Name	Description
	OnTestsetChanged	Event; fired when the user has changed the content of a test set .

See Also

[EwEMSPPlugin.UI Namespace](#)

EwEGrid

gridEmulator Constructor

Creates a new [test set configuration](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Sub New

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.gridEmulator Properties

The [gridEmulator](#) type exposes the following members.

Properties

	Name	Description
	Game	Get/set the active game to operate onto.
	Pressure	Get/set the pressure associated with a given grid row.
	Testset	Get/set the active test set to operate onto.

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.Game Property

Get/set the active [game](#) to operate onto.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Game As cGame Get Set</pre>

Property Value

Type: [cGame](#)

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.Pressure Property

Get/set the pressure associated with a given grid row.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Property Pressure (  
    iRow As Integer  
) As cPressure  
    Get  
    Set
```

Parameters

iRow

Type: [System.Int32](#)

Property Value

Type: [cPressure](#)

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.Testset Property

Get/set the active [test set](#) to operate onto.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Testset As cTestset Get Set</pre>

Property Value

Type: [cTestset](#)

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.gridEmulator Methods

The [gridEmulator](#) type exposes the following members.

Methods

	Name	Description
	FillData	Fill the grid with testset data. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)
	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellClicked	Edit handler to catch driver cell clicks. Overridden to invoke a folder browse dialog when the user clicks a Grid pressure cell. (Overrides EwEGrid.OnCellClicked(Position, ICellVirtual) .)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update testset content. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual) .)

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.FillData Method

Fill the grid with testset data.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected Overrides Sub FillData

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.FinishStyle Method

Finalizes grid formatting.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub FinishStyle

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.InitStyle Method

Initialize the grid columns and layout.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub InitStyle

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.OnCellClicked Method

Edit handler to catch driver cell clicks. Overridden to invoke a folder browse dialog when the user clicks a Grid pressure cell.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Overrides Sub OnCellClicked (  
    p As Position,  
    cell As ICellVirtual  
)
```

Parameters

p

Type: **Position**

Position that was affected.

cell

Type: **ICellVirtual**

Cell that was clicked.

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.OnCellValueChanged Method

EwEEditHandler

callback for responding to user cell value changes. Overridden to update testset content.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Protected Overrides Function OnCellValueChanged (  
    p As Position,  
    cell As ICellVirtual  
) As Boolean
```

Parameters

p

Type: **Position**

Position that was affected.

cell

Type: **ICellVirtual**

Cell that has received a new value.

Return Value

Type: [Boolean](#)

The return value is ignored by the EwEGrid framework.

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.gridEmulator Events

The [gridEmulator](#) type exposes the following members.

Events

	Name	Description
	OnTestsetChanged	Event; fired when the user has changed the content of a test set .

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.OnTestsetChanged Event

Event; fired when the user has changed the content of a [test set](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Event OnTestsetChanged As gridEmulator.OnTestsetChangedEventHandler

Value

Type: [EwEMSPPlugin.UI.gridEmulator.OnTestsetChangedEventHandler](#)

See Also

[gridEmulator Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridEmulator.OnTestsetChangedEventHandler Delegate

[Missing <summary> documentation for

"T:EwEMSPPlugin.UI.gridEmulator.OnTestsetChangedEventHandler"]

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Delegate Sub OnTestsetChangedEventHandler (  
    sender As Object,  
    t As cTestset  
)
```

Parameters

sender

Type: [System.Object](#)

t

Type: [EwEMSPPlugin.Emulator.cTestset](#)

See Also

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes Class

Outcome configuration grid; enables configuring how Ecospace predictions are fed back to MEL.

Inheritance Hierarchy

[System.Object](#)

[System.MarshalByRefObject](#)

[System.ComponentModel.Component](#)

[System.Windows.Forms.Control](#)

[System.Windows.Forms.ScrollableControl](#)

[System.Windows.Forms.Panel](#)

CustomScrollControl

GridVirtual

Grid

EwEGrid

EwEMSPPlugin.UI.gridOutcomes

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class gridOutcomes Inherits EwEGrid</pre>

The **gridOutcomes** type exposes the following members.

Constructors

	Name	Description
	gridOutcomes	Creates a new outcome configuration grid .

Properties

	Name	Description
	ItemIndex	Get/set the ordinal number of the cOutcome represented in the grid row indicated by <i>iRow</i>
	Output	Get/set the output to configure.
	Shell	Get/set the MSP shell to operate onto.

Methods

	Name	Description
	FillData	Fill the grid with outcome configurations. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)

	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update the configuration of the selected outcome. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual).)

Events

	Name	Description
	OnMappingsChanged	Event; fired when the user has changed a cOutcome configuration.

See Also

[EwEMSPPlugin.UI Namespace](#)

EwEGrid

gridOutcomes Constructor

Creates a new [outcome configuration grid](#).

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB		
Public	Sub	New

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.gridOutcomes Properties

The [gridOutcomes](#) type exposes the following members.

Properties

	Name	Description
	ItemIndex	Get/set the ordinal number of the cOutcome represented in the grid row indicated by <i>iRow</i>
	Output	Get/set the output to configure.
	Shell	Get/set the MSP shell to operate onto.

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.ItemIndex Property

Get/set the ordinal number of the [cOutcome](#) represented in the grid row indicated by *iRow*

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Protected Property ItemIndex (iRow As Integer) As Integer Get Set</pre>

Parameters

iRow

Type: [System.Int32](#)

The row to get/set the outcome index for.

Return Value

Type: [Int32](#)

The index, or 0 if an error occurred.

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.Output Property

Get/set the [output](#) to configure.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Output As cOutcome Get Set</pre>

Property Value

Type: [cOutcome](#)

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.Shell Property

Get/set the [MSP shell](#) to operate onto.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Shell As cEwEShell Get Set</pre>

Property Value

Type: [cEwEShell](#)

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.gridOutcomes Methods

The [gridOutcomes](#) type exposes the following members.

Methods

	Name	Description
	FillData	Fill the grid with outcome configurations. (Overrides EwEGrid.FillData().)
	FinishStyle	Finalizes grid formatting. (Overrides EwEGrid.FinishStyle().)
	InitStyle	Initialize the grid columns and layout. (Overrides EwEGrid.InitStyle().)
	OnCellValueChanged	EwEEditHandler callback for responding to user cell value changes. Overridden to update the configuration of the selected outcome. (Overrides EwEGrid.OnCellValueChanged(Position, ICellVirtual).)

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.FillData Method

Fill the grid with outcome configurations.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub FillData

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.FinishStyle Method

Finalizes grid formatting.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub FinishStyle

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.InitStyle Method

Initialize the grid columns and layout.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected Overrides Sub InitStyle

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.OnCellValueChanged Method

EwEEditHandler

callback for responding to user cell value changes. Overridden to update the configuration of the selected outcome.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Protected Overrides Function OnCellValueChanged (  
    p As Position,  
    cell As ICellVirtual  
) As Boolean
```

Parameters

p

Type: **Position**

Position that was affected.

cell

Type: **ICellVirtual**

Cell that has received a new value.

Return Value

Type: [Boolean](#)

The return value is ignored by the EwEGrid framework.

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.gridOutcomes Events

The [gridOutcomes](#) type exposes the following members.

Events

	Name	Description
	OnMappingsChanged	Event; fired when the user has changed a cOutcome configuration.

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.OnMappingsChanged Event

Event; fired when the user has changed a [cOutcome](#) configuration.

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Event OnMappingsChanged As gridOutcomes.OnMappingsChangedEventHandler

Value

Type: [EwEMSPPlugin.UI.gridOutcomes.OnMappingsChangedEventHandler](#)

See Also

[gridOutcomes Class](#)

[EwEMSPPlugin.UI Namespace](#)

gridOutcomes.OnMappingsChangedEventHandler Delegate

[Missing <summary> documentation for

"T:EwEMSPPlugin.UI.gridOutcomes.OnMappingsChangedEventHandler"]

Namespace: [EwEMSPPlugin.UI](#)

Assembly: EwEMSPPlugin (in EwEMSPPlugin.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Delegate Sub OnMappingsChangedEventHandler (  
    sender As gridOutcomes  
)
```

Parameters

sender

Type: [EwEMSPPlugin.UI.gridOutcomes](#)

See Also

[EwEMSPPlugin.UI Namespace](#)

EwEShell Namespaces

Maritime Strategic Planning Challenge 2050 foundation classes for connecting MEL (MSP - EwE link) and the Ecopath with Ecosim modeling approach

Namespaces

Namespace	Description
EwEShell	Maritime Strategic Planning Challenge 2050 foundation classes for connecting MEL (MSP - EwE link) and the Ecopath with Ecosim modeling approach
EwEShell.IO	Input / Output classes for EwEShell

EwEShell Namespace

Maritime Strategic Planning Challenge 2050 foundation classes for connecting MEL (MSP - EwE link) and the Ecospace with Ecosim modeling approach

Classes

	Class	Description
	cDriver	Base driver class for inserting MSP pressure data into Ecospace input variables.
	cEcospaceController	Single-threaded Ecospace execution controller for MSP game connectivity. All control functions are blocking for the purpose of MEL, waiting for the threaded Ecospace model to complete a requested operation.
	cEffortDriver	Driver for inserting MSP pressure data into the effort multiplier of a single Ecospace fleet .
	cEnvironmentalDriver	Driver for inserting MSP pressure data into a single Ecospace environmental driver map.
	cEwEShell	MEL contact point for interacting with Ecospace.
	cGame	Container for a single game (model + its configuration) in the MSP software.
	cGrid	Data for a single map in MSP.
	cHabitatDriver	Driver for inserting MSP pressure data into the map of a single Ecospace Ecospace habitat .
	cJSONGameConfig	JSON file serialization class that reads a game configuration.
	cMELException	Boink.
	cMPADriver	Driver for inserting MSP pressure data into the map of a single Ecospace Marine Protected Area .
	cOutcome	Descriptor of the content of a single ecological outcome layer in the MSP game. This class describes which Ecospace output layers need to be grouped together into a single MSP outcome layer.
	cPressure	Pressure data derived from MSP game play actions to impact the Ecospace model.
	cScalar	Data for a single value in MSP.
	cShellData	Data container for a list of games.

Interfaces

	Interface	Description
	IMELItem	Interface for defining data items in the MEL - EwE complex.

Enumerations

	Enumeration	Description
	cOutcome.eLayerType	Outcome layers categories.

 cOutcome.eMSPDiversityIndex	Initially a mirror of eDiversityIndexType , this list can include diversity indices not natively computed by EwE.
 cPressure.eDataTypes	Enumerated type, defining the types of data that a pressure supports.

cDriver Class

Base driver class for inserting MSP pressure data into Ecospace input variables.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cDriver

[EwEShell.cEffortDriver](#)

[EwEShell.cEnvironmentalDriver](#)

[EwEShell.cHabitatDriver](#)

[EwEShell.cMPADriver](#)

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public MustInherit Class cDriver Implements IMELItem</pre>

The **cDriver** type exposes the following members.

Constructors

	Name	Description
	cDriver	Creates a new driver .

Properties

	Name	Description
	Name	Get/set the name of the driver.

Methods

	Name	Description
	Apply	Apply a MSP driver value to Ecospace
	DataType	Data support identifier; override to state which pressure data type this driver can be connected to.
	ToString	String representing this driver. (Overrides Object.ToString() .)
	ValueID	Driver identifier; override to provide the unique EwE object identifier for the driver. This ID is needed for persistent pressure - driver mapping storage.

Fields

	Name	Description
	m_core	The cCore to connect to.
	m_game	The cGame to connect to.

See Also

[EwEShell Namespace](#)

cDriver Constructor

Creates a new [driver](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    core As cCore,  
    game As cGame,  
    strName As String  
)
```

Parameters

core

Type: **cCore**

The **cCore** to connect to.

game

Type: [EwEShell.cGame](#)

The [cGame](#) to connect to.

strName

Type: [System.String](#)

Name of the new driver.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.cDriver Properties

The [cDriver](#) type exposes the following members.

Properties

	Name	Description
	Name	Get/set the name of the driver.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.Name Property

Get/set the name of the driver.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.cDriver Methods

The [cDriver](#) type exposes the following members.

Methods

	Name	Description
	Apply	Apply a MSP driver value to Ecospace
	DataType	Data support identifier; override to state which pressure data type this driver can be connected to.
	ToString	String representing this driver. (Overrides Object.ToString() .)
	ValueID	Driver identifier; override to provide the unique EwE object identifier for the driver. This ID is needed for persistent pressure - driver mapping storage.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.Apply Method

Apply a MSP driver value to Ecospace

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public MustOverride Function Apply (  
    pressure As cPressure,  
    Optional data As cEcospaceDataStructures = Nothing,  
    Optional multiplier As Double = 1  
) As Boolean
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The MEL-derived pressure value to apply to the driver.

data (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace data structures to apply the driver to. If not provided, the driver data must be applied through **ICoreInputOutput** interfaces instead.

multiplier (Optional)

Type: [System.Double](#)

Return Value

Type: [Boolean](#)

True if successful.

Remarks

When used in the EwE user interface and MEL Emulator, data will be applied to **core I/O classes** to provide user feedback to test data changes. The MSP game will operate directly on the **cEcospaceDataStructures** for speed reasons, where no UI synchronizing is needed.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.DataType Method

Data support identifier; override to state which [pressure data type](#) this driver can be connected to.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public MustOverride Function DataType As cPressure.eDataTypes
```

Return Value

Type: [cPressure.eDataTypes](#)

The pressure type.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.ToString Method

String representing this driver.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ToString As String</code>

Return Value

Type: [String](#)

A [String](#) that represents this driver.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.ValueID Method

Driver identifier; override to provide the unique EwE **object identifier** for the driver. This ID is needed for persistent pressure - driver mapping storage.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public MustOverride Function ValueID As String</pre>

Return Value

Type: [String](#)

The unique ID.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.cDriver Fields

The [cDriver](#) type exposes the following members.

Fields

	Name	Description
	m_core	The cCore to connect to.
	m_game	The cGame to connect to.

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.m_core Field

The **cCore** to connect to.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected m_core As cCore

Field Value

Type: **cCore**

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cDriver.m_game Field

The [cGame](#) to connect to.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Protected m_game As cGame

Field Value

Type: [cGame](#)

See Also

[cDriver Class](#)

[EwEShell Namespace](#)

cEcospaceController Class

Single-threaded Ecospace execution controller for MSP game connectivity. All control functions are blocking for the purpose of MEL, waiting for the threaded Ecospace model to complete a requested operation.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cEcospaceController

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEcospaceController Implements IEcospaceInitializedPlugin</pre>

The **cEcospaceController** type exposes the following members.

Constructors

	Name	Description
	cEcospaceController	Initializes a new instance of the cEcospaceController class.

Properties

	Name	Description
	Author	Returns the author of this plug-in point.
	Contact	Returns the contact information of this plug-in point.
	Description	Returns the description of this plug-in point.
	IsSaveOutput	Get/set whether outcome grids need to be saved to file.
	Name	Returns the unique name for this plug-in point.
	OutputPath	Get/set the output location for saved outputs .

Methods

	Name	Description
	Continue	Perform a single Ecospace time step.
	EcospaceInitialized	Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. Implemented to receive the Ecospace data structures .
	Initialize	Initialize the plugin.

 IsRunning	State flag, returns whether Ecospace is running.
 IsStopping	State flag, returns whether Ecospace is stopping.
 Start	Start an Ecospace run.
 Stop	Terminate an Ecospace run.

Remarks

This class is implemented as a plug-in to obtain access to the **Ecospace datastructures**.

See Also

[EwEShell Namespace](#)

cEcospaceController Constructor

Initializes a new instance of the [cEcospaceController](#) class.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Sub New

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.cEcospaceController Properties

The [cEcospaceController](#) type exposes the following members.

Properties

	Name	Description
	Author	Returns the author of this plug-in point.
	Contact	Returns the contact information of this plug-in point.
	Description	Returns the description of this plug-in point.
	IsSaveOutput	Get/set whether outcome grids need to be saved to file.
	Name	Returns the unique name for this plug-in point.
	OutputPath	Get/set the output location for saved outputs .

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEospaceController.Author Property

Returns the author of this plug-in point.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Author As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Author

See Also

[cEospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.Contact Property

Returns the contact information of this plug-in point.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Contact As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Contact

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.Description Property

Returns the description of this plug-in point.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Description As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Description

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.IsSaveOutput Property

Get/set whether outcome grids need to be saved to file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property IsSaveOutput As Boolean Get Set</pre>

Return Value

Type: [Boolean](#)

True if outputs need saving to file.

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEospaceController.Name Property

Returns the unique name for this plug-in point.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Name As String Get</pre>

Property Value

Type: [String](#)

Implements

IPlugin.Name

See Also

[cEospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.OutputPath Property

Get/set the output location for [saved outputs](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property OutputPath As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.cEcospaceController Methods

The [cEcospaceController](#) type exposes the following members.

Methods

	Name	Description
	Continue	Perform a single Ecospace time step.
	EcospaceInitialized	Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. Implemented to receive the Ecospace data structures .
	Initialize	Initialize the plugin.
	IsRunning	State flag, returns whether Ecospace is running.
	IsStopping	State flag, returns whether Ecospace is stopping.
	Start	Start an Ecospace run.
	Stop	Terminate an Ecospace run.

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.Continue Method

Perform a single Ecospace time step.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub Continue (  
    pressures As cPressure(),  
    outcomes As cGrid()  
)
```

Parameters

pressures

Type: [EwEShell.cPressure\[\]](#)

The pressures to apply to the time step.

outcomes

Type: [EwEShell.cGrid\[\]](#)

The outcomes to fill during the time step.

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

[\[!:Start\(cPressure\(\), cGrid\(\)\)\]](#)

[cEcospaceController.Stop\(\)](#)

cEcospaceController.EcospaceInitialized Method

Plug-in point that is called when Ecospace has loaded a new scenario, is initialized, and is ready to be used. Implemented to receive the **Ecospace data structures**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub EcospaceInitialized (  
    EcospaceDatastructures As Object  
)
```

Parameters

EcospaceDatastructures

Type: [System.Object](#)

The ecospace datastructures that just received new scenario data.

Implements

IEcospaceInitializedPlugin.EcospaceInitialized(Object)

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.Initialize Method

Initialize the plugin.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Initialize (  
    core As Object  
)
```

Parameters

core

Type: [System.Object](#)

The core this plugin is initialized for.

Implements

IPlugin.Initialize(Object)

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.IsRunning Method

State flag, returns whether Ecospace is running.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function IsRunning As Boolean

Return Value

Type: [Boolean](#)

True if running.

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEcospaceController.IsStopping Method

State flag, returns whether Ecospace is stopping.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function IsStopping As Boolean
```

Return Value

Type: [Boolean](#)

True if stopped.

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

cEospaceController.Start Method

Start an Eospace run.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub Start (  
    game As cGame  
)
```

Parameters

game

Type: [EwEShell.cGame](#)

The [cGame](#) to launch Eospace for.

Remarks

Note that this only prepares the threads needed to run. Eospace will only start executing when Continue is called."/>

See Also

[cEospaceController Class](#)

[EwEShell Namespace](#)

[cEospaceController.Continue\(cPressure\[\],cGrid\[\]\)](#)

[cEospaceController.Stop\(\)](#)

cEcospaceController.Stop Method

Terminate an Ecospace run.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Stop
```

See Also

[cEcospaceController Class](#)

[EwEShell Namespace](#)

[cEcospaceController.Continue\(cPressure\[\],cGrid\[\]\)](#)

[cEcospaceController.Start\(cGame\)](#)

cEffortDriver Class

Driver for inserting MSP pressure data into the **effort multiplier** of a single **Ecospace fleet**.

Inheritance Hierarchy

[System.Object](#)

[EwEShell.cDriver](#)

EwEShell.cEffortDriver

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB	
Public Class	cEffortDriver
Inherits	cDriver

The **cEffortDriver** type exposes the following members.

Constructors

	Name	Description
	cEffortDriver	Create a new cEffortDriver to drive the Ecospace effort multiplier of a single fleet.

Properties

	Name	Description
	StartValue	Returns the effort multiplier configured in the base Ecospace model.

Methods

	Name	Description
	Apply	Applies the specified fishing effort multiplier. (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by scalar data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace fleet . (Overrides cDriver.ValueID() .)

See Also

[EwEShell Namespace](#)

cEffortDriver Constructor

Create a new [cEffortDriver](#) to drive the **Ecospace effort multiplier** of a single fleet.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    core As cCore,  
    game As cGame,  
    fleet As cEcopathFleetInput  
)
```

Parameters

core

Type: **cCore**

The **cCore** to connect to.

game

Type: [EwEShell.cGame](#)

The [cGame](#) to connect to.

fleet

Type: **cEcopathFleetInput**

The **fleet** this driver is connected to.

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.cEffortDriver Properties

The [cEffortDriver](#) type exposes the following members.

Properties

	Name	Description
	StartValue	Returns the effort multiplier configured in the base Ecospace model.

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.StartValue Property

Returns the effort multiplier configured in the base Ecospace model.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property StartValue As Double Get</pre>

Return Value

Type: [Double](#)

The effort multiplier configured in the base Ecospace model.

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.cEffortDriver Methods

The [cEffortDriver](#) type exposes the following members.

Methods

	Name	Description
	Apply	Applies the specified fishing effort multiplier. (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by scalar data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace fleet . (Overrides cDriver.ValueID() .)

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.Apply Method

Applies the specified fishing effort multiplier.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Function Apply (  
    pressure As cPressure,  
    Optional data As cEcospaceDataStructures = Nothing,  
    Optional multiplier As Double = 1  
) As Boolean
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The MEL-derived fishing effort multiplier value to apply to the driver.

data (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace data structures to apply pressures to.

multiplier (Optional)

Type: [System.Double](#)

The effort multiplier which translate a MEL fishing effort pressure value (0 to 1) to an Ecospace effort multiplier (0 to inf).

Return Value

Type: [Boolean](#)

Always true. Happy.

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.DataType Method

Returns that this driver can only be driven by scalar data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides Function DataType As cPressure.eDataTypes
```

Return Value

Type: [cPressure.eDataTypes](#)

The supported [pressure type](#).

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEffortDriver.ValueID Method

Returns the unique ID for the Ecospace **fleet**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ValueID As String</code>

Return Value

Type: [String](#)

The unique ID for the Ecospace **fleet**.

See Also

[cEffortDriver Class](#)

[EwEShell Namespace](#)

cEnvironmentalDriver Class

Driver for inserting MSP pressure data into a single **Ecospace environmental driver** map.

Inheritance Hierarchy

[System.Object](#)

[EwEShell.cDriver](#)

EwEShell.cEnvironmentalDriver

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEnvironmentalDriver Inherits cDriver</pre>

The **cEnvironmentalDriver** type exposes the following members.

Constructors

	Name	Description
	cEnvironmentalDriver	Create a new cMPADriver to drive the Marine Protected Area map of a given Ecospace MPA .

Methods

	Name	Description
	Apply	Insert pressure data into the environmental driver layer map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the environmental driver layer . (Overrides cDriver.ValueID() .)

See Also

[EwEShell Namespace](#)

cEnvironmentalDriver Constructor

Create a new [cMPADriver](#) to drive the **Marine Protected Area map** of a given **Ecospace MPA**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
  
Public Sub New (  
    core As cCore,  
    game As cGame,  
    layer As cEcospaceLayerDriver  
)
```

Parameters

core

Type: **cCore**

The **cCore** to connect to.

game

Type: [EwEShell.cGame](#)

The [cGame](#) to connect to.

layer

Type: **cEcospaceLayerDriver**

The **environmental driver layer** this driver is connected to.

See Also

[cEnvironmentalDriver Class](#)

[EwEShell Namespace](#)

cEnvironmentalDriver.cEnvironmentalDriver Methods

The [cEnvironmentalDriver](#) type exposes the following members.

Methods

	Name	Description
	Apply	Insert pressure data into the environmental driver layer map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the environmental driver layer . (Overrides cDriver.ValueID() .)

See Also

[cEnvironmentalDriver Class](#)

[EwEShell Namespace](#)

cEnvironmentalDriver.Apply Method

Insert pressure data into the **environmental driver layer map**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Function Apply (  
    pressure As cPressure,  
    Optional data As cEcospaceDataStructures = Nothing,  
    Optional multiplier As Double = 1  
) As Boolean
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The MEL-derived protection map value to apply to the driver.

data (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace data structures to apply pressures to.

multiplier (Optional)

Type: [System.Double](#)

Ignored.

Return Value

Type: [Boolean](#)

True if applied correctly, False if an error occurred.

Exceptions

Exception	Condition
cMELEXception	A MEL exception will be thrown if something went wrong.

See Also

[cEnvironmentalDriver Class](#)

[EwEShell Namespace](#)

cEnvironmentalDriver.DataType Method

Returns that this driver can only be driven by gridded map data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides Function DataType As cPressure.eDataTypes
```

Return Value

Type: [cPressure.eDataTypes](#)

The supported [pressure type](#).

See Also

[cEnvironmentalDriver Class](#)

[EwEShell Namespace](#)

cEnvironmentalDriver.ValueID Method

Returns the unique ID for the **environmental driver layer**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ValueID As String</code>

Return Value

Type: [String](#)

The unique ID for the **environmental driver layer**.

See Also

[cEnvironmentalDriver Class](#)

[EwEShell Namespace](#)

cEwEShell Class

MEL contact point for interacting with Ecospace.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cEwEShell

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEwEShell</pre>

The **cEwEShell** type exposes the following members.

Constructors

	Name	Description
	cEwEShell()	Stand-alone constructor when used in an EwE console environment.
	cEwEShell(cCore)	Nested constructor when used in an EwE plug-in.

Properties

	Name	Description
	Controller	Get the Ecospace controller to use.
	Core	Get the EwE core to operate onto.
	CurrentGame	Returns the currently active game , if any.
	Data	Get the EwE shell data to use.
	SaveOutcomeLocation	Get/set the directory where outcomes should be saved.
	SaveOutcomes	Get/set whether outcomes should be saved to disk.

Methods

	Name	Description
	Configuration(String, List(cScalar))	Load a new game and validate its configuration against the expectations of MEL.
	Configuration(String, String, Int32, Double, Double, Double, Int32, Int32, ICollection(cPressure), ICollection(cGrid), List(cScalar), Double)	Load a new game and validate its configuration against the expectations of MEL.

 ConfigurationFromFile	Load the game configuration from a JSON file.
 LoadConfiguration()	Load game configurations embedded within a EwE model.
 LoadConfiguration(String, String)	Save the game configuration from a XML file.
 OnChanged	Update the current configuration in the core, ready for saving. This will dirty the core for saving.
 SaveConfiguration	Save the game configuration to a XML file.
 Startup	MEL API call to start a game.
 Terminate	Stop a game.
 Tick	Run a single time step.

See Also

[EwEShell Namespace](#)

cEwEShell Constructor

Overload List

	Name	Description
	cEwEShell()	Stand-alone constructor when used in an EwE console environment.
	cEwEShell(cCore)	Nested constructor when used in an EwE plug-in.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell Constructor

Stand-alone constructor when used in an EwE console environment.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB		
Public	Sub	New

See Also

[cEwEShell Class](#)

[cEwEShell Overload](#)

[EwEShell Namespace](#)

cEwEShell Constructor (cCore)

Nested constructor when used in an EwE plug-in.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core, obtained from the master plug-in, to initialize EwE shell against

See Also

[cEwEShell Class](#)

[cEwEShell Overload](#)

[EwEShell Namespace](#)

cEwEShell.cEwEShell Properties

The [cEwEShell](#) type exposes the following members.

Properties

	Name	Description
	Controller	Get the Ecospace controller to use.
	Core	Get the EwE core to operate onto.
	CurrentGame	Returns the currently active game , if any.
	Data	Get the EwE shell data to use.
	SaveOutcomeLocation	Get/set the directory where outcomes should be saved.
	SaveOutcomes	Get/set whether outcomes should be saved to disk.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Controller Property

Get the [Ecospace controller](#) to use.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public ReadOnly Property Controller As cEcospaceController  
    Get
```

Property Value

Type: [cEcospaceController](#)

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Core Property

Get the EwE **core** to operate onto.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Core As cCore Get</pre>

Property Value

Type: **cCore**

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.CurrentGame Property

Returns the currently active [game](#), if any.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property CurrentGame As cGame Get</pre>

Property Value

Type: [cGame](#)

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Data Property

Get the [EwE shell data](#) to use.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Data As cShellData Get</pre>

Property Value

Type: [cShellData](#)

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.SaveOutcomeLocation Property

Get/set the directory where outcomes should be saved.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property SaveOutcomeLocation As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

[cEwEShell.SaveOutcomes](#)

cEwEShell.SaveOutcomes Property

Get/set whether outcomes should be saved to disk.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property SaveOutcomes As Boolean Get Set</pre>

Property Value

Type: [Boolean](#)

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

[cEwEShell.SaveOutcomeLocation](#)

cEwEShell.cEwEShell Methods

The [cEwEShell](#) type exposes the following members.

Methods

Name	Description
 Configuration(String, List(cScalar))	Load a new game and validate its configuration against the expectations of MEL.
 Configuration(String, String, Int32, Double, Double, Double, Int32, Int32, ICollection(cPressure), ICollection(cGrid), List(cScalar), Double)	Load a new game and validate its configuration against the expectations of MEL.
 ConfigurationFromFile	Load the game configuration from a JSON file.
 LoadConfiguration()	Load game configurations embedded within a EwE model.
 LoadConfiguration(String, String)	Save the game configuration from a XML file.
 OnChanged	Update the current configuration in the core, ready for saving. This will dirty the core for saving.
 SaveConfiguration	Save the game configuration to a XML file.
 Startup	MEL API call to start a game.
 Terminate	Stop a game.
 Tick	Run a single time step.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Configuration Method

Overload List

	Name	Description
	Configuration(String, List(cScalar))	Load a new game and validate its configuration against the expectations of MEL.
	Configuration(String, String, Int32, Double, Double, Double, Int32, Int32, ICollection(cPressure), ICollection(cGrid), List(cScalar), Double)	Load a new game and validate its configuration against the expectations of MEL.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Configuration Method (String, List(cScalar))

Load a new game and validate its configuration against the expectations of MEL.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function Configuration (  
    strJSON As String,  
    effortstartvalues As List(Of cScalar)  
) As Boolean
```

Parameters

strJSON

Type: [System.String](#)

Actual JSON game contract text (not the file!)

effortstartvalues

Type: [System.Collections.Generic.List\(cScalar\)](#)

[cScalar](#)

values with effort intensity start values in EwE

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[Configuration Overload](#)

[EwEShell Namespace](#)

[cEwEShell.Configuration Method \(String, String, Int32, Double, Double, Double, Int32, Int32, ICollection\(cPressure\), ICollection\(cGrid\), List\(cScalar\), Double\)](#)

Load a new game and validate its configuration against the expectations of MEL.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function Configuration (  
    strEwEModelFile As String,  
    mode As String,  
    timestep As Integer,  
    longitude As Double,  
    latitude As Double,  
    size As Double,  
    ncolumns As Integer,  
    nrows As Integer,  
    pressures As ICollection(Of cPressure),  
    outcomelayers As ICollection(Of cGrid),  
    effortstartvalues As List(Of cScalar),  
    Optional outcomerange As Double = 10  
) As Boolean
```

Parameters

strEwEModelFile

Type: [System.String](#)

The path or URI to the EwE model to (down)load.

mode

Type: [System.String](#)

The name of the game to load. Leaving the name empty will load the first available game.

timestep

Type: [System.Int32](#)

Check: the number of time steps per year.

longitude

Type: [System.Double](#)

Check: the model longitude origin to validate. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

latitude

Type: [System.Double](#)

Check: the model latitude origin to validate. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

size

Type: [System.Double](#)

Check: the model cell size. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

ncolumns

Type: [System.Int32](#)

Check: the number of columns in the model.

nrows

Type: [System.Int32](#)

Check: the number of rows in the model.

pressures

Type: [System.Collections.Generic.ICollection\(cPressure\)](#)

Check: the pressures that the model should support.

outcomelayers

Type: [System.Collections.Generic.ICollection\(cGrid\)](#)

Check: the outcomes that the model should support.

effortstartvalues

Type: [System.Collections.Generic.List\(cScalar\)](#)

[cScalar](#)

values with effort intensity start values in EwE

outcomerange (Optional)

Type: [System.Double](#)

Biomass / catch fluctuations for outcomes to display

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[Configuration Overload](#)

[EwEShell Namespace](#)

cEwEShell.ConfigurationFromFile Method

Load the game configuration from a JSON file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function ConfigurationFromFile (  
    strFile As String,  
    effortstartvalues As List(Of cScalar)  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The path/URI to the JSON file to (down)load.

effortstartvalues

Type: [System.Collections.Generic.List\(cScalar\)](#)

[cScalar](#)

values with effort intensity start values in EwE

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.LoadConfiguration Method

Overload List

	Name	Description
	LoadConfiguration()	Load game configurations embedded within a EwE model.
	LoadConfiguration(String, String)	Save the game configuration from a XML file.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.LoadConfiguration Method

Load game configurations embedded within a EwE model.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function LoadConfiguration As Boolean

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[LoadConfiguration Overload](#)

[EwEShell Namespace](#)

cEwEShell.LoadConfiguration Method (String, String)

Save the game configuration from a XML file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
  
Public Function LoadConfiguration (  
    strFile As String,  
    strMode As String  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The game configuration XML file.

strMode

Type: [System.String](#)

the game mode to use.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[LoadConfiguration Overload](#)

[EwEShell Namespace](#)

cEwEShell.OnChanged Method

Update the current configuration in the core, ready for saving. This will **dirty the core** for saving.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub OnChanged
```

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.SaveConfiguration Method

Save the game configuration to a XML file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Function SaveConfiguration (strFile As String) As Boolean</pre>

Parameters

strFile

Type: [System.String](#)

The game configuration XML file.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Startup Method

MEL API call to start a game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Startup
```

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Terminate Method

Stop a game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Terminate
```

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cEwEShell.Tick Method

Run a single time step.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub Tick (  
    pressures As List(Of cPressure),  
    outcomelayers As List(Of cGrid)  
)
```

Parameters

pressures

Type: [System.Collections.Generic.List\(cPressure\)](#)

[Missing <param name="pressures"/> documentation for "M:EwEShell.cEwEShell.Tick(System.Collections.Generic.List{EwEShell.cPressure},System.Collections.Generic.List{EwEShell.cGrid})"]

outcomelayers

Type: [System.Collections.Generic.List\(cGrid\)](#)

[Missing <param name="outcomelayers"/> documentation for "M:EwEShell.cEwEShell.Tick(System.Collections.Generic.List{EwEShell.cPressure},System.Collections.Generic.List{EwEShell.cGrid})"]

See Also

[cEwEShell Class](#)

[EwEShell Namespace](#)

cGame Class

Container for a single game (model + its configuration) in the MSP software.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cGame

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cGame Implements IMELItem</pre>

The **cGame** type exposes the following members.

Constructors

	Name	Description
	cGame	Create a new cGame .

Properties

	Name	Description
	Author	Get/set the game author information.
	CalculateIndicators	Get/set whether Ecospace must execute expensive trophic level and biodiversity indicator calculations.
	Contact	Get/set the game contact information.
	Description	Get/set the game description.
	Driver	Returns the driver connected to a specific pressure.
	Drivers	Returns the list of drivers available in the Ecospace scenario for receiving pressures.
	EcosimID	Get/set the DBID of the Ecosim scenario to load for the game.
	EcospaceID	Get/set the DBID of Ecospace scenario to load for the game.
	MPACellClosureRatio	Get/set how much of a cell needs to be covered by protection to close the cell for fishing. This value is expressed as a cell area ratio [0, 1].
	Multiplier	Returns the driver multiplier for to a specific pressure.
	Name	Get/set the unique name for the game.
	OutcomeRange	Get/set the max scale of outcome content to MEL. By default this value is 10, or one order of magnitude.

 Output	Find the output that describes how a specific outcome grid should be populated.
 Outputs	Returns the list of output configurations defined in the game.
 Pressure	Return a single pressure definitions by name.
 Pressures	Returns the list of pressures definitions in the game.
 RunYears	Get/set the number of years to run Ecospace for in the MSP game. This should be long enough to allow a full MSP run to complete, but otherwise will not cause major memory constraints.
 SpinupYears	Get/set the number of spin-up years for initializing Ecospace.
 Version	Get/set the game version.

Methods

Name	Description
 Add(cOutcome)	Add a cOutcome to the game.
 Add(cPressure)	Add a cPressure to the game.
 AddDefaultPressures	Create default pressures and outputs.
 ApplyPressures	Apply pressure layers to the running Ecospace scenario.
 EcosimScenario	Helper method, returns the actual cEcoSimScenario for the currently configured EcosimID
 EcospaceScenario	Helper method, returns the actual cEcospaceScenario for the currently configured EcospaceID
 EffortStartValues	Returns the effort start values configured in the game.
 Load	Prepare the game. Load sim and space, run space spin-up.
 LoadOutcomes	Load all outcomes from the current time step results.
 NumConnectedDrivers	Returns the number of drivers that are configured to receive pressure data.
 Remove(cOutcome)	Remove a cOutcome from the game.
 Remove(cPressure)	Remove a cPressure from the game.
 ToString	Returns an informative string summarizing the game configuration. (Overrides Object.ToString() .)
 Validate	MEL API call, validate the game configuration against expected parameters.

See Also

[EwEShell Namespace](#)

cGame Constructor

Create a new [cGame](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core to operate onto.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.cGame Properties

The [cGame](#) type exposes the following members.

Properties

	Name	Description
	Author	Get/set the game author information.
	CalculateIndicators	Get/set whether Ecospace must execute expensive trophic level and biodiversity indicator calculations.
	Contact	Get/set the game contact information.
	Description	Get/set the game description.
	Driver	Returns the driver connected to a specific pressure.
	Drivers	Returns the list of drivers available in the Ecospace scenario for receiving pressures.
	EcosimID	Get/set the DBID of the Ecosim scenario to load for the game.
	EcospaceID	Get/set the DBID of Ecospace scenario to load for the game.
	MPACellClosureRatio	Get/set how much of a cell needs to be covered by protection to close the cell for fishing. This value is expressed as a cell area ratio [0, 1].
	Multiplier	Returns the driver multiplier for to a specific pressure.
	Name	Get/set the unique name for the game.
	OutcomeRange	Get/set the max scale of outcome content to MEL. By default this value is 10, or one order of magnitude.
	Output	Find the output that describes how a specific outcome grid should be populated.
	Outputs	Returns the list of output configurations defined in the game.
	Pressure	Return a single pressure definitions by name.
	Pressures	Returns the list of pressures definitions in the game.
	RunYears	Get/set the number of years to run Ecospace for in the MSP game. This should be long enough to allow a full MSP run to complete, but otherwise will not cause major memory constraints.
	SpinupYears	Get/set the number of spin-up years for initializing Ecospace.
	Version	Get/set the game version.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Author Property

Get/set the game author information.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Author As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.CalculateIndicators Property

Get/set whether Ecospace must execute expensive trophic level and biodiversity indicator calculations.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property CalculateIndicators As Boolean Get Set</pre>

Property Value

Type: [Boolean](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Contact Property

Get/set the game contact information.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Contact As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Description Property

Get/set the game description.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Description As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Driver Property

Returns the driver connected to a specific pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property Driver (  
    pressure As String  
) As cDriver  
    Get  
    Set
```

Parameters

pressure

Type: [System.String](#)

The name of the pressure to find the driver for.

Return Value

Type: [cDriver](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Drivers Property

Returns the list of drivers available in the Ecospace scenario for receiving pressures.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public ReadOnly Property Drivers (  
    Optional datatype As cPressure.eDataTypes = cPressure.eDataTypes.NotSet  
) As cDriver()  
    Get
```

Parameters

datatype (Optional)

Type: [EwEShell.cPressure.eDataTypes](#)

Property Value

Type: [cDriver](#)[]

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.EcosimID Property

Get/set the **DBID** of the Ecosim scenario to load for the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property EcosimID As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.EcospaceID Property

Get/set the **DBID** of Ecospace scenario to load for the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property EcospaceID As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.MPACellClosureRatio Property

Get/set how much of a cell needs to be covered by protection to close the cell for fishing. This value is expressed as a cell area ratio [0, 1].

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property MPACellClosureRatio As Single  
    Get  
    Set
```

Property Value

Type: [Single](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Multiplier Property

Returns the driver multiplier for to a specific pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property Multiplier (  
    pressure As String  
) As Double  
    Get  
    Set
```

Parameters

pressure

Type: [System.String](#)

The name of the pressure to find the multiplier for.

Return Value

Type: [Double](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Name Property

Get/set the unique name for the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.OutcomeRange Property

Get/set the max scale of outcome content to MEL. By default this value is 10, or one order of magnitude.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property OutcomeRange As Double  
    Get  
    Set
```

Property Value

Type: [Double](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Output Property

Find the output that describes how a specific outcome grid should be populated.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public ReadOnly Property Output (  
    grid As cGrid  
) As cOutcome  
    Get
```

Parameters

grid

Type: [EwEShell.cGrid](#)

The grid to find the output for.

Return Value

Type: [cOutcome](#)

An output if configured, or nothing if the output could not be found.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Outputs Property

Returns the list of output configurations defined in the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Outputs As ICollection(Of cOutcome) Get</pre>

Property Value

Type: [ICollection\(cOutcome\)](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Pressure Property

Return a single pressure definitions by name.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public ReadOnly Property Pressure (  
    strName As String  
) As cPressure  
    Get
```

Parameters

strName

Type: [System.String](#)

The name of the pressure to retrieve.

Property Value

Type: [cPressure](#)

Remarks

The name search is not case sensitive.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Pressures Property

Returns the list of pressures definitions in the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Pressures As ICollection(Of cPressure) Get</pre>

Property Value

Type: [ICollection\(cPressure\)](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.RunYears Property

Get/set the number of years to run Ecospace for in the MSP game. This should be long enough to allow a full MSP run to complete, but otherwise will not cause major memory constraints.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property RunYears As Integer  
    Get  
    Set
```

Property Value

Type: [Int32](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.SpinupYears Property

Get/set the number of spin-up years for initializing Ecospace.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property SpinupYears As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Version Property

Get/set the game version.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Version As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.cGame Methods

The [cGame](#) type exposes the following members.

Methods

	Name	Description
	Add(cOutcome)	Add a cOutcome to the game.
	Add(cPressure)	Add a cPressure to the game.
	AddDefaultPressures	Create default pressures and outputs.
	ApplyPressures	Apply pressure layers to the running Ecospace scenario.
	EcosimScenario	Helper method, returns the actual cEcoSimScenario for the currently configured EcosimID
	EcospaceScenario	Helper method, returns the actual cEcospaceScenario for the currently configured EcospaceID
	EffortStartValues	Returns the effort start values configured in the game.
	Load	Prepare the game. Load sim and space, run space spin-up.
	LoadOutcomes	Load all outcomes from the current time step results.
	NumConnectedDrivers	Returns the number of drivers that are configured to receive pressure data.
	Remove(cOutcome)	Remove a cOutcome from the game.
	Remove(cPressure)	Remove a cPressure from the game.
	ToString	Returns an informative string summarizing the game configuration. (Overrides Object.ToString() .)
	Validate	MEL API call, validate the game configuration against expected parameters.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Add Method

Overload List

	Name	Description
	Add(cOutcome)	Add a cOutcome to the game.
	Add(cPressure)	Add a cPressure to the game.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Add Method (cOutcome)

Add a [cOutcome](#) to the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Add (  
    output As cOutcome  
)
```

Parameters

output

Type: [EwEShell.cOutcome](#)

The [cOutcome](#) to add.

See Also

[cGame Class](#)

[Add Overload](#)

[EwEShell Namespace](#)

cGame.Add Method (cPressure)

Add a [cPressure](#) to the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Add (  
    pressure As cPressure  
)
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The [cPressure](#) to add.

See Also

[cGame Class](#)

[Add Overload](#)

[EwEShell Namespace](#)

cGame.AddDefaultPressures Method

Create default pressures and outputs.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub AddDefaultPressures
```

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.ApplyPressures Method

Apply pressure layers to the running Ecospace scenario.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function ApplyPressures (  
    pressurelayers As cPressure(),  
    Optional spaceDS As cEcospaceDataStructures = Nothing  
) As Boolean
```

Parameters

pressurelayers

Type: [EwEShell.cPressure\[\]](#)

The pressures to apply.

spaceDS (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace datastructures to directly apply pressures to.

Return Value

Type: [Boolean](#)

True if successful.

Exceptions

Exception	Condition
cMELEXception	A MEL exception will be thrown if something went wrong.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.EcosimScenario Method

Helper method, returns the actual **cEcoSimScenario** for the currently configured [EcosimID](#)

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function EcosimScenario As cEcoSimScenario
```

Return Value

Type: **cEcoSimScenario**

A scenario, or nothing if this could not be found.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.EcospaceScenario Method

Helper method, returns the actual **cEcospaceScenario** for the currently configured [EcospaceID](#)

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function EcospaceScenario As cEcospaceScenario
```

Return Value

Type: **cEcospaceScenario**

A scenario, or nothing if this could not be found.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.EffortStartValues Method

Returns the effort start values configured in the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function EffortStartValues As ICollection(Of cScalar)
```

Return Value

Type: [ICollection\(cScalar\)](#)

A collection of Scalars (fleet name, value).

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Load Method

Prepare the game. Load sim and space, run space spin-up.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function Load As Boolean

Return Value

Type: [Boolean](#)

[Missing <returns> documentation for "M:EwEShell.cGame.Load"]

Exceptions

Exception	Condition
cMELErrorException	A MEL exception will be thrown if something went wrong.

Remarks

True if successful.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.LoadOutcomes Method

Load all outcomes from the current time step results.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function LoadOutcomes (  
    outcomelayers As cGrid(),  
    results As cEcospaceTimestep  
) As Boolean
```

Parameters

outcomelayers

Type: [EwEShell.cGrid\[\]](#)

The outcome layers to populate.

results

Type: **cEcospaceTimestep**

Ecospace time step data to populate the outcomes from.

Return Value

Type: [Boolean](#)

True if successful.

Exceptions

Exception	Condition
cMELEXception	A MEL exception will be thrown if something went wrong.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.NumConnectedDrivers Method

Returns the number of drivers that are configured to receive pressure data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function NumConnectedDrivers As Integer

Return Value

Type: [Int32](#)

The number of drivers that are configured to receive pressure data.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Remove Method

Overload List

	Name	Description
	Remove(cOutcome)	Remove a cOutcome from the game.
	Remove(cPressure)	Remove a cPressure from the game.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Remove Method (cOutcome)

Remove a [cOutcome](#) from the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Remove (  
    output As cOutcome  
)
```

Parameters

output

Type: [EwEShell.cOutcome](#)

The [cOutcome](#) to remove.

See Also

[cGame Class](#)

[Remove Overload](#)

[EwEShell Namespace](#)

cGame.Remove Method (cPressure)

Remove a [cPressure](#) from the game.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Remove (  
    pressure As cPressure  
)
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The [cPressure](#) to remove.

See Also

[cGame Class](#)

[Remove Overload](#)

[EwEShell Namespace](#)

cGame.ToString Method

Returns an informative string summarizing the game configuration.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ToString As String</code>

Return Value

Type: [String](#)

Minime

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGame.Validate Method

MEL API call, validate the game configuration against expected parameters.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Function Validate (  
    timestep As Integer,  
    longitude As Double,  
    latitude As Double,  
    size As Double,  
    ncolumns As Integer,  
    nrows As Integer,  
    pressures As ICollection(Of cPressure),  
    outcomelayers As ICollection(Of cGrid)  
) As Boolean
```

Parameters

timestep

Type: [System.Int32](#)

Check: the number of time steps per year.

longitude

Type: [System.Double](#)

Check: the model longitude origin to validate. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

latitude

Type: [System.Double](#)

Check: the model latitude origin to validate. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

size

Type: [System.Double](#)

Check: the model cell size. Ignored for now because of slight numerical precision differences between MSP, MEL and EwE.

ncolumns

Type: [System.Int32](#)

Check: the number of columns in the model.

nrows

Type: [System.Int32](#)

Check: the number of rows in the model.

pressures

Type: [System.Collections.Generic.ICollection\(cPressure\)](#)

EwE tools for MSP - API documentation reference

Check: the pressures that the model should support.

outcomelayers

Type: [System.Collections.Generic.ICollection\(cGrid\)](#)

Check: the outcomes that the model should support.

Return Value

Type: [Boolean](#)

True if the game validated.

Exceptions

Exception	Condition
cMEException	A MEL exception will be thrown if something went wrong.

See Also

[cGame Class](#)

[EwEShell Namespace](#)

cGrid Class

Data for a single map in MSP.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cGrid

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cGrid Implements IMELItem</pre>

The **cGrid** type exposes the following members.

Constructors

	Name	Description
	cGrid	Creates a new cGrid .

Properties

	Name	Description
	Cell	Get the zero-base indexed data for the grid, dimensioned as (width x height)
	Height	Get the height or number of rows in the grid.
	IsValid	Get/set whether the content of the grid is valid.
	Max	Get/set the highest value in the output grid.
	Mean	Get/set the mean value in the output grid.
	Min	Get/set the lowest value in the output grid.
	Name	Get/set the name of the map.
	NumValueCells	Get/set the number of non-zero value cells in the output.
	Units	Get/set the units in which grid values are expressed.
	Width	Get/set the width or number of columns in the grid.

Methods

	Name	Description
	Clear	Erase the content of the grid.
	Load	Loads the map from file.

EwE tools for MSP - API documentation reference

	Save	Saves the map to file.
---	----------------------	------------------------

See Also

[EwEShell Namespace](#)

cGrid Constructor

Creates a new [cGrid](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    name As String,  
    iWidth As Integer,  
    iHeight As Integer,  
    Optional data As Double(,) = Nothing  
)
```

Parameters

name

Type: [System.String](#)

The name for the grid.

iWidth

Type: [System.Int32](#)

The width or number of columns in the grid.

iHeight

Type: [System.Int32](#)

The height or number of rows in the grid.

data (Optional)

Type: [System.Double\[,\]](#)

Data for the grid, if any.

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.cGrid Properties

The [cGrid](#) type exposes the following members.

Properties

	Name	Description
	Cell	Get the zero-base indexed data for the grid, dimensioned as (width x height)
	Height	Get the height or number of rows in the grid.
	IsValid	Get/set whether the content of the grid is valid.
	Max	Get/set the highest value in the output grid.
	Mean	Get/set the mean value in the output grid.
	Min	Get/set the lowest value in the output grid.
	Name	Get/set the name of the map.
	NumValueCells	Get/set the number of non-zero value cells in the output.
	Units	Get/set the units in which grid values are expressed.
	Width	Get/set the width or number of columns in the grid.

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Cell Property

Get the zero-base indexed data for the grid, dimensioned as (width x height)

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Cell As Double(,) Get</pre>

Property Value

Type: [Double](#)(,)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Height Property

Get the height or number of rows in the grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Height As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.IsValid Property

Get/set whether the content of the grid is valid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property IsValid As Boolean Get Set</pre>

Property Value

Type: [Boolean](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Max Property

Get/set the highest value in the output grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Max As Double Get Set</pre>

Property Value

Type: [Double](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Mean Property

Get/set the mean value in the output grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Mean As Double Get Set</pre>

Property Value

Type: [Double](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Min Property

Get/set the lowest value in the output grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Min As Double Get Set</pre>

Property Value

Type: [Double](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Name Property

Get/set the name of the map.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.NumValueCells Property

Get/set the number of non-zero value cells in the output.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property NumValueCells As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Units Property

Get/set the units in which grid values are expressed.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Units As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Width Property

Get/set the width or number of columns in the grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Width As Integer Get Set</pre>

Property Value

Type: [Int32](#)

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.cGrid Methods

The [cGrid](#) type exposes the following members.

Methods

	Name	Description
	Clear	Erase the content of the grid.
	Load	Loads the map from file.
	Save	Saves the map to file.

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Clear Method

Erase the content of the grid.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Clear
```

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Load Method

Loads the map from file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Load (  
    strFile As String,  
    core As cCore  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The file name to load from. File names with the .asc extension, as well as non-compressed image files, are supported

core

Type: **cCore**

The core that provides spatial contextual information.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cGrid.Save Method

Saves the map to file.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Save (  
    strFile As String,  
    core As cCore  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The file name to save to. Only file names with the .asc extension are supported

core

Type: **cCore**

The core that provides spatial contextual information.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cGrid Class](#)

[EwEShell Namespace](#)

cHabitatDriver Class

Driver for inserting MSP pressure data into the **map** of a single Ecospace **Ecospace habitat**.

Inheritance Hierarchy

[System.Object](#)

[EwEShell.cDriver](#)

EwEShell.cHabitatDriver

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Class cHabitatDriver
    Inherits cDriver
```

The **cHabitatDriver** type exposes the following members.

Constructors

	Name	Description
	cHabitatDriver	Create a new cHabitatDriver to drive the habitat map of a given Ecospace habitat .

Methods

	Name	Description
	Apply	Insert pressure data into the habitat map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace habitat . (Overrides cDriver.ValueID() .)

See Also

[EwEShell Namespace](#)

cHabitatDriver Constructor

Create a new [cHabitatDriver](#) to drive the **habitat map** of a given **Ecospace habitat**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    core As cCore,  
    game As cGame,  
    hab As cEcospaceHabitat  
)
```

Parameters

core

Type: **cCore**

The **cCore** to connect to.

game

Type: [EwEShell.cGame](#)

The [cGame](#) to connect to.

hab

Type: **cEcospaceHabitat**

The **habitat** this driver is connected to.

See Also

[cHabitatDriver Class](#)

[EwEShell Namespace](#)

cHabitatDriver.cHabitatDriver Methods

The [cHabitatDriver](#) type exposes the following members.

Methods

	Name	Description
	Apply	Insert pressure data into the habitat map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace habitat . (Overrides cDriver.ValueID() .)

See Also

[cHabitatDriver Class](#)

[EwEShell Namespace](#)

cHabitatDriver.Apply Method

Insert pressure data into the **habitat map**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Function Apply (  
    pressure As cPressure,  
    Optional data As cEcospaceDataStructures = Nothing,  
    Optional multiplier As Double = 1  
) As Boolean
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The MEL-derived habitat map value to apply to the driver.

data (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace data structures to apply pressures to.

multiplier (Optional)

Type: [System.Double](#)

Ignored.

Return Value

Type: [Boolean](#)

True if applied correctly, False if an error occurred.

Exceptions

Exception	Condition
cMELEXception	A MEL exception will be thrown if something went wrong.

See Also

[cHabitatDriver Class](#)

[EwEShell Namespace](#)

cHabitatDriver.DataType Method

Returns that this driver can only be driven by gridded map data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides Function DataType As cPressure.eDataTypes
```

Return Value

Type: [cPressure.eDataTypes](#)

The supported [pressure type](#).

See Also

[cHabitatDriver Class](#)

[EwEShell Namespace](#)

cHabitatDriver.ValueID Method

Returns the unique ID for the Ecospace **habitat**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ValueID As String</code>

Return Value

Type: [String](#)

The unique ID for the Ecospace **habitat**.

See Also

[cHabitatDriver Class](#)

[EwEShell Namespace](#)

cJSONGameConfig Class

JSON file serialization class that reads a game configuration.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cJSONGameConfig

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cJSONGameConfig</pre>

The **cJSONGameConfig** type exposes the following members.

Constructors

	Name	Description
	cJSONGameConfig	Create a new cJSONGameConfig .

Properties

	Name	Description
	CellSize	Get the game area cell size provided in the JSON text.
	EwEModelFile	Get the EwE model file provided in the JSON text.
	Latitude	Get the game area latitude (northern edge) provided in the JSON text.
	Longitude	Get the game area longitude (western edge) provided in the JSON text.
	Mode	Get the EwE game mode provided in the JSON text.
	NumColumns	Get the game area number of columns provided in the JSON text.
	NumRows	Get the game area number of rows provided in the JSON text.
	OutcomeRange	Get the game outcome range in the JSON text.
	Outcomes	Get the outcome definitions provided in the JSON text.
	Pressures	Get the pressure definitions provided in the JSON text.
	Timestep	Get the number of time steps per year provided in the JSON text.

Methods

	Name	Description
	Load	Extracts game configuration from the specified JSON text.
	LoadFile	Loads a JSON file and extracts game configuration.

EwE tools for MSP - API documentation reference

See Also

[EwEShell Namespace](#)

cJSONGameConfig Constructor

Create a new [cJSONGameConfig](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Public Sub New

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.cJSONGameConfig Properties

The [cJSONGameConfig](#) type exposes the following members.

Properties

	Name	Description
	CellSize	Get the game area cell size provided in the JSON text.
	EwEModelFile	Get the EwE model file provided in the JSON text.
	Latitude	Get the game area latitude (northern edge) provided in the JSON text.
	Longitude	Get the game area longitude (western edge) provided in the JSON text.
	Mode	Get the EwE game mode provided in the JSON text.
	NumColumns	Get the game area number of columns provided in the JSON text.
	NumRows	Get the game area number of rows provided in the JSON text.
	OutcomeRange	Get the game outcome range in the JSON text.
	Outcomes	Get the outcome definitions provided in the JSON text.
	Pressures	Get the pressure definitions provided in the JSON text.
	Timestep	Get the number of time steps per year provided in the JSON text.

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.CellSize Property

Get the game area cell size provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property CellSize As Double Get</pre>

Property Value

Type: [Double](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.EwEModelFile Property

Get the EwE model file provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property EwEModelFile As String Get</pre>

Property Value

Type: [String](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Latitude Property

Get the game area latitude (northern edge) provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Latitude As Double Get</pre>

Property Value

Type: [Double](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Longitude Property

Get the game area longitude (western edge) provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Longitude As Double Get</pre>

Property Value

Type: [Double](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Mode Property

Get the EwE game mode provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Mode As String Get</pre>

Property Value

Type: [String](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.NumColumns Property

Get the game area number of columns provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property NumColumns As Integer Get</pre>

Property Value

Type: [Int32](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.NumRows Property

Get the game area number of rows provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property NumRows As Integer Get</pre>

Property Value

Type: [Int32](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.OutcomeRange Property

Get the game outcome range in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property OutcomeRange As Double Get</pre>

Property Value

Type: [Double](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Outcomes Property

Get the outcome definitions provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre><JsonIgnoreAttribute> Public ReadOnly Property Outcomes As cGrid() Get</pre>

Property Value

Type: [cGrid\[\]](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Pressures Property

Get the pressure definitions provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
<JsonIgnoreAttribute>  
Public ReadOnly Property Pressures As cPressure()  
    Get
```

Property Value

Type: [cPressure](#)[]

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Timestep Property

Get the number of time steps per year provided in the JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Timestep As Integer Get</pre>

Property Value

Type: [Int32](#)

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.cJSONGameConfig Methods

The [cJSONGameConfig](#) type exposes the following members.

Methods

	Name	Description
	Load	Extracts game configuration from the specified JSON text.
	LoadFile	Loads a JSON file and extracts game configuration.

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.Load Method

Extracts game configuration from the specified JSON text.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Load (  
    strJSON As String  
) As Boolean
```

Parameters

strJSON

Type: [System.String](#)

The text to extract game information from.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cJSONGameConfig.LoadFile Method

Loads a JSON file and extracts game configuration.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Function LoadFile (strJSONfile As String) As Boolean</pre>

Parameters

strJSONfile

Type: [System.String](#)

The JSON file to read and extract game information from.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cJSONGameConfig Class](#)

[EwEShell Namespace](#)

cMELEException Class

Boink.

Inheritance Hierarchy

[System.Object](#)

[System.Exception](#)

EwEShell.cMELEException

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cMELEException Inherits Exception</pre>

The **cMELEException** type exposes the following members.

Constructors

	Name	Description
	cMELEException	Create a new cMELEException .

See Also

[EwEShell Namespace](#)

cMEException Constructor

Create a new [cMEException](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    strDetails As String  
)
```

Parameters

strDetails

Type: [System.String](#)

The exception details.

See Also

[cMEException Class](#)

[EwEShell Namespace](#)

cMPADriver Class

Driver for inserting MSP pressure data into the **map** of a single Ecospace **Marine Protected Area**.

Inheritance Hierarchy

[System.Object](#)

[EwEShell.cDriver](#)

EwEShell.cMPADriver

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cMPADriver Inherits cDriver</pre>

The **cMPADriver** type exposes the following members.

Constructors

	Name	Description
	cMPADriver	Create a new cMPADriver to drive the Marine Protected Area map of a given Ecospace MPA .

Methods

	Name	Description
	Apply	Insert pressure data into the MPA map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace MPA . (Overrides cDriver.ValueID() .)

See Also

[EwEShell Namespace](#)

cMPADriver Constructor

Create a new [cMPADriver](#) to drive the **Marine Protected Area map** of a given **Ecospace MPA**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    core As cCore,  
    game As cGame,  
    MPA As cEcospaceMPA  
)
```

Parameters

core

Type: **cCore**

The **cCore** to connect to.

game

Type: [EwEShell.cGame](#)

The [cGame](#) to connect to.

MPA

Type: **cEcospaceMPA**

The **MPA** this driver is connected to.

See Also

[cMPADriver Class](#)

[EwEShell Namespace](#)

cMPADriver.cMPADriver Methods

The [cMPADriver](#) type exposes the following members.

Methods

	Name	Description
	Apply	Insert pressure data into the MPA map . (Overrides cDriver.Apply(cPressure, cEcospaceDataStructures, Double) .)
	DataType	Returns that this driver can only be driven by gridded map data. (Overrides cDriver.DataType() .)
	ValueID	Returns the unique ID for the Ecospace MPA . (Overrides cDriver.ValueID() .)

See Also

[cMPADriver Class](#)

[EwEShell Namespace](#)

cMPADriver.Apply Method

Insert pressure data into the **MPA map**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Overrides Function Apply (  
    pressure As cPressure,  
    Optional data As cEcospaceDataStructures = Nothing,  
    Optional multiplier As Double = 1  
) As Boolean
```

Parameters

pressure

Type: [EwEShell.cPressure](#)

The MEL-derived protection map value to apply to the driver.

data (Optional)

Type: **cEcospaceDataStructures**

Optional Ecospace data structures to apply pressures to.

multiplier (Optional)

Type: [System.Double](#)

Ignored.

Return Value

Type: [Boolean](#)

True if applied correctly, False if an error occurred.

Exceptions

Exception	Condition
cMELEXception	A MEL exception will be thrown if something went wrong.

See Also

[cMPADriver Class](#)

[EwEShell Namespace](#)

cMPADriver.DataType Method

Returns that this driver can only be driven by gridded map data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Overrides Function DataType As cPressure.eDataTypes
```

Return Value

Type: [cPressure.eDataTypes](#)

The supported [pressure type](#).

See Also

[cMPADriver Class](#)

[EwEShell Namespace](#)

cMPADriver.ValueID Method

Returns the unique ID for the Ecospace **MPA**.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ValueID As String</code>

Return Value

Type: [String](#)

The unique ID for the Ecospace **MPA**.

See Also

[cMPADriver Class](#)

[EwEShell Namespace](#)

cOutcome Class

Descriptor of the content of a single ecological outcome layer in the MSP game. This class describes which Ecospace output layers need to be grouped together into a single MSP outcome layer.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cOutcome

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cOutcome Implements IMELItem</pre>

The **cOutcome** type exposes the following members.

Constructors

	Name	Description
	cOutcome	Constructor.

Properties

	Name	Description
	Denominator	Get/set the denominator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.
	LayerType	Returns the category of the layer.
	Name	Get/set the name of the output.
	Numerator	Get/set the numerator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.

Methods

	Name	Description
	IsConfigured	Returns if this output is configured to receive data.
	NumItems	Returns the total number of items that can be aggregated in the outcome layer.
	NumUsed	Get the number of items that will be aggregated in the outcome layer.
	Populate	Populate the actual outcome layer.
	ToString	Returns an informative string for the outcome layer configuration. (Overrides Object.ToString() .)

EwE tools for MSP - API documentation reference

See Also

[EwEShell Namespace](#)

cOutcome Constructor

Constructor.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    core As cCore,  
    strName As String,  
    type As cOutcome.eLayerType  
)
```

Parameters

core

Type: **cCore**

The core to draw information from.

strName

Type: [System.String](#)

The name of the outcome layer.

type

Type: [EwEShell.cOutcome.eLayerType](#)

The [category](#) of the layer.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.cOutcome Properties

The [cOutcome](#) type exposes the following members.

Properties

	Name	Description
	Denominator	Get/set the denominator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.
	LayerType	Returns the category of the layer.
	Name	Get/set the name of the output.
	Numerator	Get/set the numerator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.Denominator Property

Get/set the denominator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property Denominator (  
    iIndex As Integer  
) As Double  
    Get  
    Set
```

Parameters

iIndex

Type: [System.Int32](#)

The one-based index of the item to configure, which ranges from 1 to [the total number of items](#).

Property Value

Type: [Double](#)

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.LayerType Property

Returns the [category](#) of the layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property LayerType As cOutcome.eLayerType Get Set</pre>

Property Value

Type: [cOutcome.eLayerType](#)

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.Name Property

Get/set the name of the output.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.Numerator Property

Get/set the numerator weight for ecospace results for a specific item (group or fleet) for composing the outcome layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Property Numerator (  
    iIndex As Integer  
) As Double  
    Get  
    Set
```

Parameters

iIndex

Type: [System.Int32](#)

The one-based index of the item to configure, which ranges from 1 to [the total number of items](#).

Property Value

Type: [Double](#)

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.cOutcome Methods

The [cOutcome](#) type exposes the following members.

Methods

	Name	Description
	IsConfigured	Returns if this output is configured to receive data.
	NumItems	Returns the total number of items that can be aggregated in the outcome layer.
	NumUsed	Get the number of items that will be aggregated in the outcome layer.
	Populate	Populate the actual outcome layer.
	ToString	Returns an informative string for the outcome layer configuration. (Overrides Object.ToString() .)

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.IsConfigured Method

Returns if this output is configured to receive data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function IsConfigured As Boolean

Return Value

Type: [Boolean](#)

True if this instance is configured.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.NumItems Method

Returns the total number of items that can be aggregated in the outcome layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function NumItems As Integer
```

Return Value

Type: [Int32](#)

The total number of items that can be aggregated in the outcome layer.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.NumUsed Method

Get the number of items that will be aggregated in the outcome layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Function NumUsed As Integer

Return Value

Type: [Int32](#)

The total number of items that will be aggregated in the outcome layer.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.Populate Method

Populate the actual outcome layer.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
  
Public Overridable Function Populate (  
    grid As cGrid,  
    timestepdata As cEcospaceTimestep,  
    outcomerange As Double  
) As Boolean
```

Parameters

grid

Type: [EwEShell.cGrid](#)

The grid to populate.

timestepdata

Type: **cEcospaceTimestep**

The Ecospace time step data that contains the Ecospace estimates for the current time step.

outcomerange

Type: [System.Double](#)

Outcome range, specifying the magnitude of variation in Ecospace predictions reported in the outcome map.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.ToString Method

Returns an informative string for the outcome layer configuration.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Overrides Function ToString As String</code>

Return Value

Type: [String](#)

Ooooooh!

See Also

[cOutcome Class](#)

[EwEShell Namespace](#)

cOutcome.eLayerType Enumeration

Outcome layers categories.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Enumeration eLayerType

Members

Member name	Value	Description
Biomass	0	Outcome contains an Ecospace biomass distribution summary.
Catch	1	Outcome contains an Ecospace catch distribution summary.
Effort	2	Outcome contains an Ecospace fishing effort distribution summary.
Indicator	3	Outcome contains an spatially explicit ecological indicator .

See Also

[EwEShell Namespace](#)

cOutcome.eMSPDiversityIndex Enumeration

Initially a mirror of **eDiversityIndexType**, this list can include diversity indices not natively computed by EwE.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Enumeration eMSPDiversityIndex</code>

Members

Member name	Value	Description
Shannon	0	Shannon diversity indicator.
Kempton's Q	1	Kempton's Q indicator.

See Also

[EwEShell Namespace](#)

cPressure Class

Pressure data derived from MSP game play actions to impact the Ecospace model.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cPressure

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cPressure Implements IMELItem</pre>

The **cPressure** type exposes the following members.

Constructors

	Name	Description
	cPressure(String, Double)	Create a scalar pressure.
	cPressure(String, Int32, Int32, Double[,])	Create a grid pressure.

Properties

	Name	Description
	DataType	Get the type of data that a pressure supports.
	Grid	Get a reference to the cGrid wrapped by the pressure, if this is a Grid pressure.
	IsGrid	Returns if this pressure accepts grid driver data.
	IsScalar	Returns if this pressure accepts scalar driver data.
	Name	Get/set the name of the pressure.
	Scalar	Get/set the scalar value wrapped by the pressure, if this is a Scalar pressure.

Fields

	Name	Description
	m_grid	The grid data wrapped by the pressure.
	m_scalar	The scalar data wrapped by the pressure.

Remarks

In a [MSP game](#), player actions translate to pressures. This pressure data is received in cPressure classes, and are passed on to mapped [Ecospace drivers](#) to impact the Ecospace model.

EwE tools for MSP - API documentation reference

See Also

[EwEShell Namespace](#)

cPressure Constructor

Overload List

	Name	Description
	cPressure(String, Double)	Create a scalar pressure.
	cPressure(String, Int32, Int32, Double[,])	Create a grid pressure.

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure Constructor (String, Double)

Create a [scalar](#) pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    name As String,  
    scalar As Double  
)
```

Parameters

name

Type: [System.String](#)

The name of the pressure to define.

scalar

Type: [System.Double](#)

The initial data for the pressure.

See Also

[cPressure Class](#)

[cPressure Overload](#)

[EwEShell Namespace](#)

[cPressure.DataType](#)

[EwEShell.cPressure.eDataTypes](#)

[cPressure.Scalar](#)

cPressure Constructor (String, Int32, Int32, Double[,])

Create a [grid](#) pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Sub New (  
    name As String,  
    iNumColumns As Integer,  
    iNumRows As Integer,  
    Optional data As Double(,) = Nothing  
)
```

Parameters

name

Type: [System.String](#)

The name of the pressure to define.

iNumColumns

Type: [System.Int32](#)

The number of columns in the pressure grid.

iNumRows

Type: [System.Int32](#)

The number of rows in the pressure grid.

data (Optional)

Type: [System.Double\[,\]](#)

Optional initial data for the pressure.

See Also

[cPressure Class](#)

[cPressure Overload](#)

[EwEShell Namespace](#)

[cPressure.DataType](#)

[EwEShell.cPressure.eDataTypes](#)

[cPressure.Grid](#)

cPressure.cPressure Properties

The [cPressure](#) type exposes the following members.

Properties

	Name	Description
	DataType	Get the type of data that a pressure supports.
	Grid	Get a reference to the cGrid wrapped by the pressure, if this is a Grid pressure.
	IsGrid	Returns if this pressure accepts grid driver data.
	IsScalar	Returns if this pressure accepts scalar driver data.
	Name	Get/set the name of the pressure.
	Scalar	Get/set the scalar value wrapped by the pressure, if this is a Scalar pressure.

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.DataType Property

Get the [type of data](#) that a pressure supports.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property DataType As cPressure.eDataTypes Get</pre>

Property Value

Type: [cPressure.eDataTypes](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.Grid Property

Get a reference to the [cGrid](#) wrapped by the pressure, if this is a [Grid](#) pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public ReadOnly Property Grid As cGrid  
    Get
```

Return Value

Type: [cGrid](#)

The grid, or null if the pressure does not accept grid data.

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

[cPressure.IsGrid](#)

cPressure.IsGrid Property

Returns if this pressure accepts grid driver data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public ReadOnly Property IsGrid As Boolean  
    Get
```

Property Value

Type: [Boolean](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.IsScalar Property

Returns if this pressure accepts scalar driver data.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property IsScalar As Boolean Get</pre>

Property Value

Type: [Boolean](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.Name Property

Get/set the name of the pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.Scalar Property

Get/set the scalar value wrapped by the pressure, if this is a [Scalar](#) pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Scalar As Double Get Set</pre>

Return Value

Type: [Double](#)

The scalar value, or 0 if the pressure does not accept scalar data.

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

[cPressure.IsScalar](#)

cPressure.cPressure Fields

The [cPressure](#) type exposes the following members.

Fields

	Name	Description
	m_grid	The grid data wrapped by the pressure.
	m_scalar	The scalar data wrapped by the pressure.

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.m_grid Field

The grid data wrapped by the pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected m_grid **As** cGrid

Field Value

Type: [cGrid](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.m_scalar Field

The scalar data wrapped by the pressure.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

Protected m_scalar **As** cScalar

Field Value

Type: [cScalar](#)

See Also

[cPressure Class](#)

[EwEShell Namespace](#)

cPressure.eDataTypes Enumeration

Enumerated type, defining the types of data that a pressure supports.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Enumeration eDataTypes</code>

Members

Member name	Value	Description
NotSet	0	Data type has not been set yet.
Scalar	1	Pressure contains a single value.
Grid	2	Pressure contains an map of values.

See Also

[EwEShell Namespace](#)

cScalar Class

Data for a single value in MSP.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cScalar

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cScalar Implements IMELItem</pre>

The **cScalar** type exposes the following members.

Constructors

	Name	Description
	cScalar	Creates a new cScalar .

Properties

	Name	Description
	Name	Get/set the name of the scalar.
	Value	Get/set the value of the scalar.

See Also

[EwEShell Namespace](#)

cScalar Constructor

Creates a new [cScalar](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    name As String,  
    value As Double  
)
```

Parameters

name

Type: [System.String](#)

The name for the scalar.

value

Type: [System.Double](#)

The value to assign to the scalar.

See Also

[cScalar Class](#)

[EwEShell Namespace](#)

cScalar.cScalar Properties

The [cScalar](#) type exposes the following members.

Properties

	Name	Description
	Name	Get/set the name of the scalar.
	Value	Get/set the value of the scalar.

See Also

[cScalar Class](#)

[EwEShell Namespace](#)

cScalar.Name Property

Get/set the name of the scalar.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Name As String Get Set</pre>

Property Value

Type: [String](#)

Implements

[IMELItem.Name](#)

See Also

[cScalar Class](#)

[EwEShell Namespace](#)

cScalar.Value Property

Get/set the value of the scalar.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Property Value As Double Get Set</pre>

Property Value

Type: [Double](#)

See Also

[cScalar Class](#)

[EwEShell Namespace](#)

cShellData Class

Data container for a list of games.

Inheritance Hierarchy

[System.Object](#)

EwEShell.cShellData

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Class cShellData

The **cShellData** type exposes the following members.

Constructors

	Name	Description
	cShellData	Constructor.

Properties

	Name	Description
	Games	Get all available games.

Methods

	Name	Description
	Add	Add a game .
	Game	Obtain a cGame by name.
	Remove	Remove a game .

See Also

[EwEShell Namespace](#)

cShellData Constructor

Constructor.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The core to operate onto.

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.cShellData Properties

The [cShellData](#) type exposes the following members.

Properties

	Name	Description
	Games	Get all available games.

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.Games Property

Get all available games.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property Games As cGame() Get</pre>

Property Value

Type: [cGame](#)[]

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.cShellData Methods

The [cShellData](#) type exposes the following members.

Methods

	Name	Description
	Add	Add a game .
	Game	Obtain a cGame by name.
	Remove	Remove a game .

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.Add Method

Add a [game](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Add (  
    g As cGame  
)
```

Parameters

g

Type: [EwEShell.cGame](#)

The g.

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.Game Method

Obtain a [cGame](#) by name.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Game (  
    strGame As String  
) As cGame
```

Parameters

strGame

Type: [System.String](#)

The name of the game. It is.

Return Value

Type: [cGame](#)

The game with a given name, or Nothing if the game could not be found.

Remarks

To obtain the first game pass in an empty name

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

cShellData.Remove Method

Remove a [game](#).

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub Remove (  
    g As cGame  
)
```

Parameters

g

Type: [EwEShell.cGame](#)

The g.

See Also

[cShellData Class](#)

[EwEShell Namespace](#)

IMELItem Interface

Interface for defining data items in the MEL - EwE complex.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
Public Interface IMELItem

The **IMELItem** type exposes the following members.

Properties

	Name	Description
	Name	Get/set the name of the item.

See Also

[EwEShell Namespace](#)

IMELItem.IMELItem Properties

The [IMELItem](#) type exposes the following members.

Properties

	Name	Description
	Name	Get/set the name of the item.

See Also

[IMELItem Interface](#)

[EwEShell Namespace](#)

IMELItem.Name Property

Get/set the name of the item.

Namespace: [EwEShell](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Property Name As String Get Set</pre>

Property Value

Type: [String](#)

See Also

[IMELItem Interface](#)

[EwEShell Namespace](#)

EwEShell.IO Namespace

Input / Output classes for EwEShell

Classes

	Class	Description
	cEcospaceImportExportBitmap	Ecospace import/export class for accessing bitmap files. MEL uses TIF files for caching computed pressure layers, which can be read via this class.

Enumerations

	Enumeration	Description
	cEcospaceImportExportBitmap.eBands	Enumerated type, defining the colour bands contained in bitmap data.

cEcospaceImportExportBitmap Class

Ecospace import/export class for accessing bitmap files. MEL uses TIF files for caching computed pressure layers, which can be read via this class.

Inheritance Hierarchy

[System.Object](#)

EwEShell.IO.cEcospaceImportExportBitmap

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public Class cEcospaceImportExportBitmap Implements IEcospaceImportExport</pre>

The **cEcospaceImportExportBitmap** type exposes the following members.

Constructors

	Name	Description
	cEcospaceImportExportBitmap	Create a new instance.

Properties

	Name	Description
	CellSize	Get the size of a cell in the Ecospace basemap.
	InCol	Get the number of columns in the Ecospace basemap.
	InRow	Get the number of rows in the Ecospace basemap.
	NoDataValue	Get the NoData value to use.
	PosTopLeft	Get the top-left (lon, lat) location of the Ecospace basemap.
	ProjectionString	Get the WKT projection string of the Ecospace basemap.
	Value	Get/set a grid value.

Methods

	Name	Description
	Load(String)	Import the specified file with default options: the Red band is used, and bands are presumed to have 256 colour values.
	Load(String, cEcospaceImportExportBitmap.eBands, Single)	Load the map from a bitmap file.

 Save	Save the map to a bitmap file.
 ToRaster	Convert the buffered data to a ISpatialRaster for use by EwE.

See Also

[EwEShell.IO Namespace](#)

IEcospaceImportExport

cEcospaceImportExportBitmap Constructor

Create a new instance.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Sub New (  
    core As cCore  
)
```

Parameters

core

Type: **cCore**

The EwE core to operate onto.

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.cEcospaceImportExportBitmap Properties

The [cEcospaceImportExportBitmap](#) type exposes the following members.

Properties

	Name	Description
	CellSize	Get the size of a cell in the Ecospace basemap.
	InCol	Get the number of columns in the Ecospace basemap.
	InRow	Get the number of rows in the Ecospace basemap.
	NoDataValue	Get the NoData value to use.
	PosTopLeft	Get the top-left (lon, lat) location of the Ecospace basemap.
	ProjectionString	Get the WKT projection string of the Ecospace basemap.
	Value	Get/set a grid value.

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.CellSize Property

Get the size of a cell in the Ecospace basemap.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property CellSize As Double Get</pre>

Property Value

Type: [Double](#)

Implements

IEcospaceImportExport.CellSize

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.InCol Property

Get the number of columns in the Ecospace basemap.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property InCol As Integer Get</pre>

Property Value

Type: [Int32](#)

Implements

IEcospaceImportExport.InCol

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.InRow Property

Get the number of rows in the Ecospace basemap.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property InRow As Integer Get</pre>

Property Value

Type: [Int32](#)

Implements

IEcospaceImportExport.InRow

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.NoDataValue Property

Get the NoData value to use.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property NoDataValue As Double Get</pre>

Property Value

Type: [Double](#)

Implements

IEcospaceImportExport.NoDataValue

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.PosTopLeft Property

Get the top-left (lon, lat) location of the Ecospace basemap.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property PosTopLeft As PointF Get</pre>

Property Value

Type: [PointF](#)

Implements

IEcospaceImportExport.PosTopLeft

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.ProjectionString Property

Get the WKT projection string of the Ecospace basemap.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<pre>Public ReadOnly Property ProjectionString As String Get</pre>

Property Value

Type: [String](#)

Implements

IEcospaceImportExport.ProjectionString

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.Value Property

Get/set a grid value.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
Public Property Value (  
    iRow As Integer,  
    iCol As Integer,  
    Optional strField As String = ""  
) As Object  
    Get  
    Set
```

Parameters

iRow

Type: [System.Int32](#)

One-based row index to access a value for.

iCol

Type: [System.Int32](#)

One-based column index to access a value for.

strField (Optional)

Type: [System.String](#)

Optional field to access a value for.

Property Value

Type: [Object](#)

Implements

IEcospaceImportExport.Value

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.cEcospaceImportExportBitmap Methods

The [cEcospaceImportExportBitmap](#) type exposes the following members.

Methods

	Name	Description
	Load(String)	Import the specified file with default options: the Red band is used, and bands are presumed to have 256 colour values.
	Load(String, cEcospaceImportExportBitmap.eBands, Single)	Load the map from a bitmap file.
	Save	Save the map to a bitmap file.
	ToRaster	Convert the buffered data to a ISpatialRaster for use by EwE.

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.Load Method

Overload List

	Name	Description
	Load(String)	Import the specified file with default options: the Red band is used, and bands are presumed to have 256 colour values.
	Load(String, cEcospaceImportExportBitmap.eBands, Single)	Load the map from a bitmap file.

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.Load Method (String)

Import the specified file with default options: the [Red](#) band is used, and bands are presumed to have 256 colour values.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Load (  
    strFile As String  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The file to import.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEcospaceImportExportBitmap Class](#)

[Load Overload](#)

[EwEShell.IO Namespace](#)

[cEcospaceImportExportBitmap.Load\(String, cEcospaceImportExportBitmap.eBands, Single\)](#)

cEcospaceImportExportBitmap.Load Method (String, cEcospaceImportExportBitmap.eBands, Single)

Load the map from a bitmap file.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

```
VB  
  
Public Function Load (  
    strFile As String,  
    band As cEcospaceImportExportBitmap.eBands,  
    Optional sScale As Single = 0.00390625F  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The file to import.

band

Type: [EwEShell.IO.cEcospaceImportExportBitmap.eBands](#)

The [band](#) to read.

sScale (Optional)

Type: [System.Single](#)

The conversion factor to convert pixel band colours to data values.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEcospaceImportExportBitmap Class](#)

[Load Overload](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.Save Method

Save the map to a bitmap file.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function Save (  
    strFile As String  
) As Boolean
```

Parameters

strFile

Type: [System.String](#)

The file to import.

Return Value

Type: [Boolean](#)

True if successful.

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

cEcospaceImportExportBitmap.ToRaster Method

Convert the buffered data to a **ISpatialRaster** for use by EwE.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB

```
Public Function ToRaster (  
    Optional strField As String = ""  
) As ISpatialRaster
```

Parameters

strField (Optional)

Type: [System.String](#)

Ignored.

Return Value

Type: **ISpatialRaster**

A populated raster.

Implements

IEcospaceImportExport.ToRaster(String)

See Also

[cEcospaceImportExportBitmap Class](#)

[EwEShell.IO Namespace](#)

ISpatialRaster

cEcospaceImportExportBitmap.eBands Enumeration

Enumerated type, defining the colour bands contained in bitmap data.

Namespace: [EwEShell.IO](#)

Assembly: EwEShell (in EwEShell.dll) Version: 1.0.0.0 (1.0.0.0)

Syntax

VB
<code>Public Enumeration eBands</code>

Members

	Member name	Value	Description
	Red	0	Red pixel value
	Green	1	Green pixel value
	Blue	2	Blue pixel value

Remarks

Alpha channel access is not supported.

See Also

[EwEShell.IO Namespace](#)